

Decentralized Federated Learning with Non-IID Data: Challenges, Trends, and Future Opportunities

WU-CHUN CHUNG, Information and Computer Engineering, Chung Yuan Christian University, Taoyuan City, Taiwan

CHAO-AI LO, Information and Computer Engineering, Chung Yuan Christian University, Taoyuan, Taiwan

YAN-HUI LIN, Information and Computer Engineering, Chung Yuan Christian University, Taoyuan City, Taiwan

ZHI-HAO CHEN, Information and Computer Engineering, Chung Yuan Christian University, Taoyuan City, Taiwan

CHE-LUN HUNG, Institute of Biomedical Informatics, National Yang Ming Chiao Tung University, Taipei City, Taiwan and Health Innovation Center, National Yang Ming Chiao Tung University, Taipei City, Taiwan

As artificial intelligence and machine learning advance, increasing privacy concerns and regulatory constraints have limited cross-border data sharing for traditional model training. Federated Learning (FL) offers a privacy-preserving approach by enabling distributed training without exposing raw data. However, FL faces significant challenges, particularly when dealing with Non-Independent and Identically Distributed (Non-IID) data, which results in inconsistent model performance. Moreover, relying on a central server also raises reliability and scalability issues. Decentralized Federated Learning (DFL) eliminates the central server, thereby fostering more robust and scalable collaboration. Despite the growing interest in DFL, a comprehensive review focusing on Non-IID challenges remains scarce. This article presents a Systematic Literature Review (SLR) of existing research on DFL under Non-IID settings. Studies were retrieved from six major academic publishers and categorized into four pillars: architecture, topology, optimization, and security. The SLR review provides insights into current trends and systematically summarizes real-world applications, commonly used datasets, and neural network models. This article also examines prevalent methods for conducting Non-IID experiments and evaluating performance metrics. By providing a structured analysis of the literature, experimental setups, and evaluation practices, this survey highlights key trends, uncovers research gaps, and proposes future directions for advancing DFL in Non-IID environments.

This work was partially supported by the National Science and Technology Council of Taiwan under Grant no: NSTC 112-2221-E-033-024-MY3.

Authors' Contact Information: Wu-Chun Chung (corresponding author), Information and Computer Engineering, Chung Yuan Christian University, Taoyuan City, Taiwan; e-mail: wcchung@cycu.edu.tw; Chao-Ai Lo, Information and Computer Engineering, Chung Yuan Christian University, Taoyuan, Taiwan; e-mail: jauai0119@cycu.org.tw; Yan-Hui Lin, Information and Computer Engineering, Chung Yuan Christian University, Taoyuan City, Taiwan; e-mail: phil890310@cycu.org.tw; Zhi-Hao Chen, Information and Computer Engineering, Chung Yuan Christian University, Taoyuan City, Taiwan; e-mail: zhchen@cycu.org.tw; Che-Lun Hung (corresponding author), Institute of Biomedical Informatics, National Yang Ming Chiao Tung University, Taipei City, Taiwan and Health Innovation Center, National Yang Ming Chiao Tung University, Taipei City, Taiwan; e-mail: clhung@nycu.edu.tw.



This work is licensed under a Creative Commons Attribution 4.0 International License.

© 2026 Copyright held by the owner/author(s).

ACM 0360-0300/2026/02-ART192

<https://doi.org/10.1145/3785657>

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Distributed computing methodologies**; • **Computer systems organization** → **Distributed architectures**;

Additional Key Words and Phrases: Federated learning, decentralized federated learning, Non-IID, systematic literature review

ACM Reference Format:

Wu-Chun Chung, Chao-Ai Lo, Yan-Hui Lin, Zhi-Hao Chen, and Che-Lun Hung. 2026. Decentralized Federated Learning with Non-IID Data: Challenges, Trends, and Future Opportunities. *ACM Comput. Surv.* 58, 8, Article 192 (February 2026), 41 pages. <https://doi.org/10.1145/3785657>

1 Introduction

Artificial Intelligence (AI) is booming with the improvement of hardware computing power. **Machine Learning (ML)** [1] is a significant AI technology that enables machines to autonomously learn data features and improve training efficiency. However, traditional ML adopts a centralized learning method. The client data is uploaded to a centralized server for consolidation and model training. When the model training is completed, the client requests the model from the server through network communication. Applying the centralized approach for ML can train general models for all clients, but lacks effective communication and privacy protection. At the same time, the difficulty of ML is to ensure the accuracy and practical effectiveness of the model when the dataset is small or contains malicious data. Therefore, model training in a distributed manner has emerged to address these issues, as known as distributed ML [2].

Distributed ML improves the impact of centralized data on model learning while enhancing training efficiency. However, distributed ML does not address the privacy and security issues. Privacy and security of personal data have become important issues, as stated in the **General Data Protection Regulation (GDPR)** [3] of the European Union. In response to these regulations and challenges, Google introduced a new ML method in 2016 called **Federated Learning (FL)** [4]. FL is a type of distributed ML method that maintains data privacy by keeping data locally. Clients utilize their local datasets for model training. Once the training process of the client is completed, the model parameters are sent to a central server for aggregation. In other words, only the extracted features are transferred to the remote server during the training process, thereby ensuring the privacy of raw data. FL not only protects the privacy of client data, but also reduces the size of bulk data communications.

Challenges in FL. On the other hand, many existing works assume that the data distribution across all clients is **Independent and Identically Distributed (IID)** [4], which means that different clients have the same data distribution. However, the data distribution between each client may vary in real cases, known as the problem of **Non-Independent and Identically Distributed (Non-IID)** data. When the clients exhibit Non-IID data, more iterations are required to achieve convergence in model aggregation. Therefore, FL requires more communication rounds to maintain a certain level of accuracy in Non-IID data scenarios [4]. NIID Benchmark [5] also reveals that the data distribution in Non-IID data is closer to the real-world situation and uses a unified standard to evaluate state-of-the-art algorithms. NIID concludes that Non-IID data has a significant impact on all algorithms and no algorithm can be generally applicable to all scenarios. Based on the literature review, the existing works on **Decentralized Federated Learning (DFL)** mainly focus on privacy security and blockchain applications. The previous surveys [6, 7] indicate that the problem of Non-IID data is worthy of discussion in FL, which is even more severe in DFL.

While a considerable amount of research and studies have been presented in the past years, FL is still in a developmental stage with some unresolved issues. Traditional **Centralized Federated**

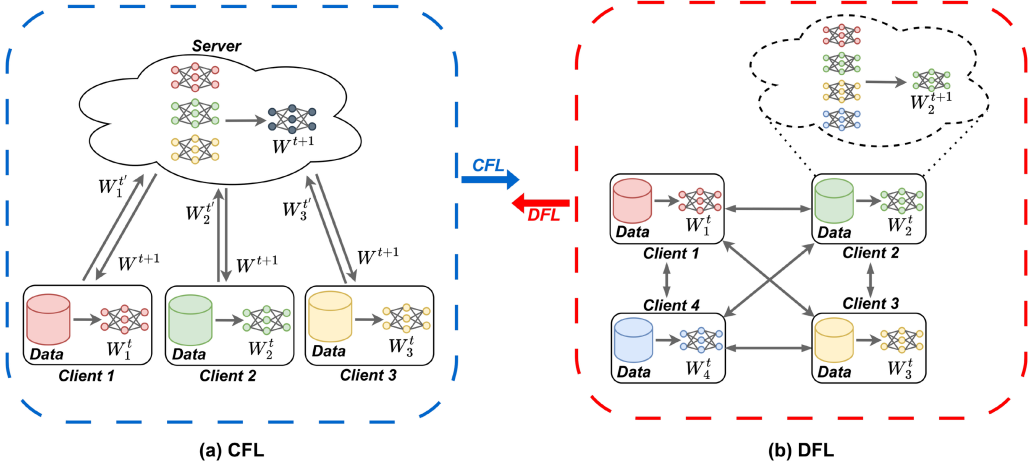


Fig. 1. Illustration of typical FL architectures.

Learning (CFL) presents various potential problems, such as a single point of failure, high communication costs, and issues related to node trust. To address these issues, the paradigm shift from the centralized architecture with a parameter server to a decentralized topology without the central server is referred to as DFL. DFL alleviates the risk of a single point of failure in CFL while reducing communication costs through efficient transmission algorithms. Moreover, DFL allows clients to choose communication peers based on neighborhood information rather than uploading models to a central server in a traditional way. The differences between CFL and DFL are presented in Figure 1. Figure 1(a) illustrates the CFL architecture, where participating clients train the model using local datasets. These clients upload model parameters to the central server for model aggregation and update the parameters for the next round of model training. Figure 1(b) depicts the basic structure of DFL, where participating clients communicate with each other based on a fully connected network. During the training process, the model parameters are exchanged with each other through a **Peer-to-Peer (P2P)** approach. Model aggregation is performed at each client to produce new model parameters for the next round of model training instead of using a central server.

Existing solutions in DFL. DFL has significant advancements in military, healthcare, and **Internet of Things (IoT)** applications [8]. However, new challenges, such as network communication, model optimization, data distribution, and security trust, are also exposed to the research. Recent research has proposed an innovative solution for FL to address the issue of excessive reliance on central servers by integrating blockchain technology. The previous surveys [9–11] conducted a detailed integration of the research on FL combined with blockchain. The combination of FL and blockchain can transform CFL into a decentralized architecture. The advantage lies in making the learning process more transparent while enhancing client trust. In addition to blockchain, the latest surveys on DFL [8, 12] involved comprehensive research aspects such as architecture, network structure, communication methods, and security. Their work reviewed and compared various DFL methods and functionalities currently available in the literature.

However, in Non-IID data scenarios, model convergence may be challenging due to the lack of a central server in DFL, resulting in reduced overall performance. Existing studies on DFL have not thoroughly explored issues related to data distribution. To address the research gaps, this article delves into the relevant studies on Non-IID data in existing DFL methodologies. Specifically, a **Systematic Literature Review (SLR)** [13, 14] method is adopted to examine all relevant studies in six scientific databases from 2018 to 2024. After five layers of filtering, a total of 268 published

papers were retrieved, and 93 studies were identified after the filtering process. The topics are discussed and categorized into four major taxonomies: architecture, topology, optimization, and security. Based on the taxonomies, this article summarizes current solutions and experimental methodologies. Finally, further research potentials are proposed to guide future directions.

The main contributions of this work are listed as follows:

– **Comprehensive SLR on DFL with Non-IID data**

This article presents the first SLR focused on the challenges and solutions of Non-IID data within DFL systems. Through rigorous screening and classification of 93 peer-reviewed studies from six major academic and scientific databases, the review provides a structured overview of the field so as to fill a critical gap left by prior surveys that predominantly focused on CFL or only single aspects of DFL.

– **Multi-dimensional taxonomy and in-depth analysis**

This article introduces a comprehensive taxonomy of four key pillars: architecture, topology, optimization, and security, to categorize and analyze the current research trends. Our work also summarizes key details from each reviewed study, such as application scenarios, datasets, models, Non-IID data generation methods, and evaluation metrics. The multi-dimensional framework allows a thorough examination of DFL methods, experimental designs, and the relationship between system design and Non-IID data challenges so as to provide new insights for researchers and practitioners.

– **Summary of findings and future directions**

This article clarifies prevailing trends and research gaps in current DFL studies. Our work also identifies unresolved issues and proposes promising future directions so as to serve as a concrete reference for advancing the robustness and applicability of DFL in real-world Non-IID environments.

Structure of this article. The remainder of this article is organized as follows. Section 2 provides a preliminary overview of FL and other related surveys. Section 3 introduces the SLR methodology to process relevant studies. Section 4 summarizes the progress of current research based on the exploration of DFL. Section 5 details the relevant experiments and statistical analysis of Non-IID data in research works. Sections 6 and 7 cover valuable findings, challenges, and concluding remarks for future research.

2 Background and Related Works

FL is an advanced ML paradigm known for its unique approach of preserving the raw data of a client locally. In traditional centralized ML, data from multiple devices is uploaded to a central server for training a unified model. FL only uploads model parameters or gradients to the central server for aggregation during the training process. Only uploading parameters or gradients not only reduces the size of transmitted data, but also ensures the privacy and security of client information. Aggregating client-specific models into a global model is a crucial concern in FL, leading to the development of the Federated Averaging (FedAvg) algorithm [4].

2.1 Background of FL Training

In conventional centralized ML, the objective is to minimize the loss function, as formulated below:

$$\min_{\omega \in \mathbb{R}^d} f(\omega), \quad f(\omega) := \frac{1}{K} \sum_{i=1}^K f_i(\omega) \quad (1)$$

where K is the total number of clients. Let $f(\omega)$ represent the objective function, and $f_i(\omega)$ is defined as the loss function for client i . $f_i(\omega) = l(x_i, y_i; \omega)$, indicating the prediction loss for the

ALGORITHM 1: FedAvg in CFL**Input:** all client K indexed by i ; local batch size B ; local epochs E ; learning rate η ; selection probability C **Output:** the final global model ω^T initialize global model ω^0 **for** each round $t = 1, 2, \dots, T$ **do**

Determined for each round the number of participated clients

 $m \leftarrow \max(C \cdot K, 1)$

Randomly select participated clients

 $S^t \leftarrow$ (random set of m clients) **for** each client $i \in S^t$ **in parallel do** $\omega_i^{t'} \leftarrow \text{ClientLocalTrain}(i, \omega^t)$ **end for**

Aggregating model

 $\omega^{t+1} \leftarrow \sum_{i=1}^K \frac{n_i}{n} \omega_i^{t'}$ **end for****function** ClientLocalTrain(i, ω^t): batched $\mathcal{B} \leftarrow$ (split D_i into batches of size B) **for** local epoch $e = 1, 2, \dots, E$ **do** **for** batch $b \in \mathcal{B}$ **do** $\omega^{t'} \leftarrow \omega^t - \eta \nabla L(\omega^t; b)$ **end for** **end for** return $\omega^{t'}$

example (x_i, y_i) using model parameters ω . However, this approach fails to consider the impact of heterogeneous data. Suboptimal convergence may occur in the case of a highly uneven distribution of data among clients.

In FL, P_i is employed as the index number for participating clients in the training. n represents the total data size, and n_i denotes the data size of the client i . When a particular client possesses a larger quantity of data, their influence in the aggregation process becomes more significant. In each round, the loss function $f(\omega)$ of the global model is updated based on the weighted average of participating clients. The ultimate optimization function for FL is expressed as follows:

$$f(\omega) = \sum_{i=1}^K \frac{n_i}{n} F_i(\omega), \text{ where } F_i(\omega) = \frac{1}{n_i} \sum_{k \in P_i} f_k(\omega) \quad (2)$$

Algorithm 1 describes the training process of FL, which is divided into two parts: the server is responsible for aggregating the global model, and the client is responsible for local training. During the training process, the hyperparameters are specified initially including the total number of rounds, batch size, local epochs, and learning rate. The server uses the global model ω^0 as the initial model. Before each training round, the number of clients participating in the training is determined and clients are randomly selected, as shown in lines 5 to 8. Lines 17 to 24 indicate that the client selected for training divides its local datasets into batches of size B and performs local training using gradient descent. The computation formula is as follows:

$$\omega^{t'} \leftarrow \omega^t - \eta \nabla L(\omega^t; b) \quad (3)$$

where, η is the learning rate, $\nabla L(\omega^t; b)$ represents the gradient of the loss function L with respect to parameter ω^t , and b represents the sample data. Through gradient descent, the model gradually

ALGORITHM 2: FedAvg in DFL**Input :** local batch size B ; local epochs E ; learning rate η ; set of neighbors $\mathcal{N}_i$ **Output :** the final models ω_i^T aggregated by all client

Each client model is exchanged over a fully connected network.

initialize client model $\omega_i^t \leftarrow \omega_i^0$ **for** each training round $t = 1, 2, \dots, T$ **do**

Local train

 batched $\mathcal{B} \leftarrow$ (split D_i into batches of size B) **for** local epoch $e = 1, 2, \dots, E$ **do** **for** batch $b \in \mathcal{B}$ **do**

$$\omega_i^{t'} \leftarrow \omega_i^t - \eta \nabla L(\omega_i^t; b)$$

end for **end for**

#Exchange local models

 Send local model $\omega_i^{t'}$ to all neighborhood client $j \in \mathcal{N}_i$ Set of model parameter $S = \{\omega_i^{t'}, \omega_j^{t'} | j \in \mathcal{N}_i\}$

Parameter aggregation

$$\omega_i^{t+1} \leftarrow \frac{1}{|S|} \sum_{s=1}^{|S|} \omega_s^{t'}$$

end for

moves in the direction of minimizing the loss. This update process is repeated until the loss function reaches its minimum or converges to a value close to the minimum. When local training is completed, the client sends the model $\omega_i^{t'}$ back to the server and waits for the next round of training. Once the models returned by participating clients are collected, the aggregation step is initiated, as shown in line 15. The aggregated model then serves as the update model for the next round. This process is repeated until the predefined number of rounds is reached. Finally, a global consensus model is obtained so that the FL training is complete.

The foundational algorithms of DFL eliminate the original central server and enable each client to independently conduct model training and aggregation. Algorithm 2 provides a brief overview of the training process in a fully connected DFL network, where clients communicate with their neighbors through P2P communication. Lines 4 to 10 indicate that each client uses local raw data to train a local model $\omega_i^{t'}$ at the beginning of each training round. Based on the defined neighbor set $\mathcal{N}_i$, model parameters are exchanged with neighbors as shown in lines 14 to 16. Subsequently, as per line 18, the client aggregates the received neighbor model $\omega_j^{t'}$ with the local model to calculate the averaged model ω_i^{t+1} . Ultimately, the aggregated model becomes the update model for the next round.

Assuming that the data distribution in the environment is an IID data scenario, FedAvg can intuitively aggregate models from individual clients. Ultimately, the server can be integrated into a generalized model. However, this assumption is controversial in real-world scenarios because all clients may not have the same data distribution. Consequently, the Non-IID data problem arises. Since the direction of model convergence may vary significantly for each client, the global model may drift during the training.

As illustrated in Figure 2, the model divergence occurs after several rounds [4]. The challenge is to achieve model convergence even in the Non-IID data scenario. Algorithm 2 presents a general version based on FedAvg to tackle some problems in the centralized architecture. However, unresolved challenges posed by Non-IID data still arise, which may become more apparent in a decentralized framework. Accordingly, figuring out how the research works on Non-IID data has progressed under the DFL framework is the objective of this work.

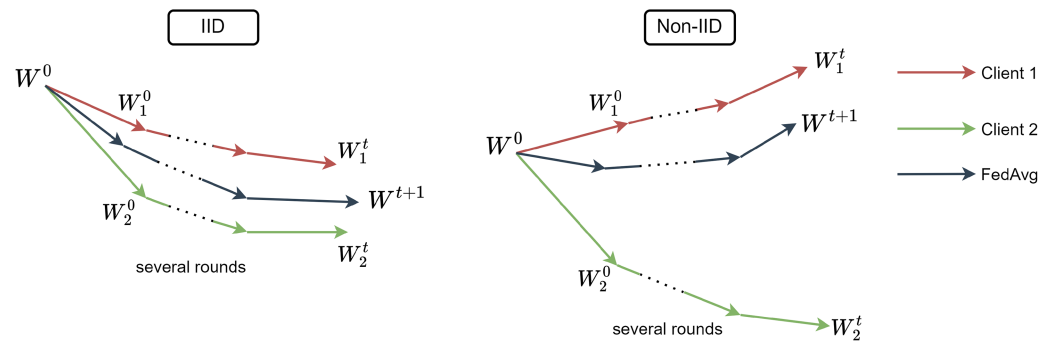


Fig. 2. Illustration of IID and Non-IID data in FL.

Table 1. Comparative Overview of Surveys on Non-IID CFL and DFL

Ref.	Year	Type of FL	Focused Topic	Type of survey	Application	Datasets	Training Models	Performance Metrics
[6]	2021	CFL	Non-IID	Unstructured review				
[7]	2024	CFL	Non-IID	Unstructured review				
[21]	2022	CFL	Non-IID	Unstructured review		✓		
[9]	2022	DFL	Security	Unstructured review	✓			
[20]	2023	DFL	Security	SLR				
[10]	2024	DFL	Security	Unstructured review	✓			
[11]	2024	DFL	Security	Unstructured review				
[26]	2023	DFL	Optimization	Unstructured review	✓	✓	✓	
[8]	2023	DFL	Architecture	Unstructured review	✓	✓	✓	✓
[12]	2024	DFL	Architecture	Unstructured review	✓			
Our	2025	DFL	Non-IID	SLR	✓	✓	✓	✓

2.2 Comparisons with Existing Survey Papers

Many studies [15–17, 172] have addressed similar research objectives to delve into the challenges and issues faced by FL through thorough investigations. The extensive investigations conducted in these studies aim at providing a comprehensive understanding of the existing challenges in FL and to suggest corresponding solutions. As research progresses, attention shifts toward specific topics in FL and has continued to increase after 2021. To better understand the research trends during this time, this article selects the most relevant survey papers for comparison, as shown in Table 1. The table lists the research surveys related to CFL, Non-IID data, and DFL. Detailed discussions and research gaps are presented in the Appendix section.

The main distinction of our work from previous studies is the adoption of an SLR framework to investigate Non-IID data issues within DFL comprehensively. The approach enables a structured and in-depth analysis of current challenges, methodologies, and experimental practices in the DFL context, which have not been systematically explored in earlier literature reviews. To address this gap, this article addresses four central pillars: architecture, topology, optimization, and security. Relevant studies from six major databases, published between 2021 and 2024, were rigorously selected. After a multi-stage filtering process, 93 representative empirical and methodological studies were included for multi-level classification and statistical analysis. Furthermore, our work explicitly examines DFL in practical application scenarios,

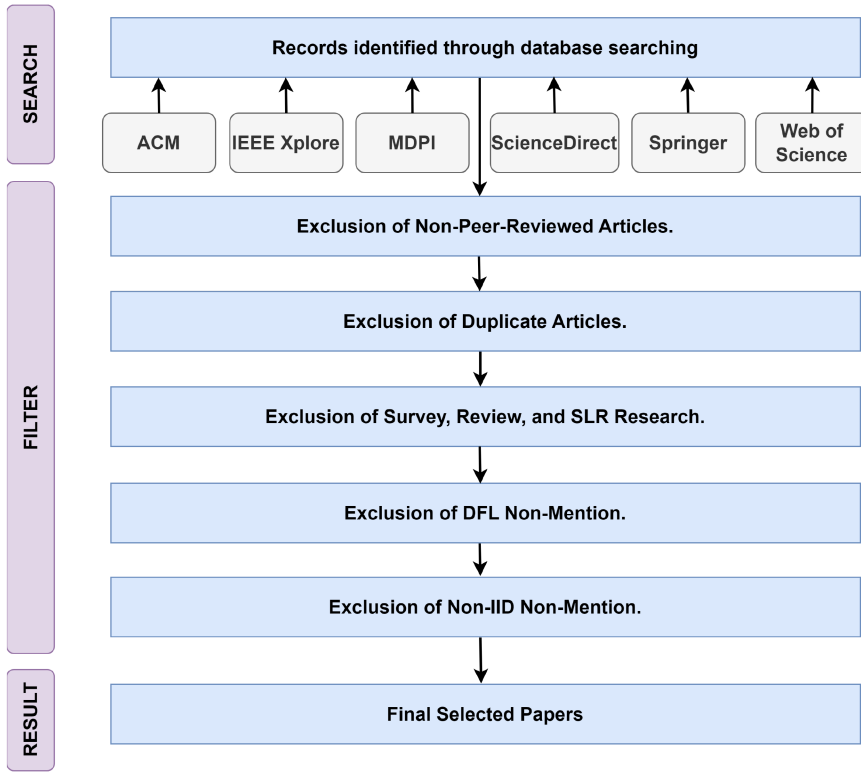


Fig. 3. The detailed SLR process.

experimental datasets, neural network models, Non-IID experimental designs, and performance evaluation metrics, while identifying current challenges and future research directions in the DFL field.

3 Methodology of Systematic Literature Review

SLR is a systematic synthesis technique used to investigate, identify, and evaluate research questions. The review methodology requires a thorough review of relevant research surrounding a specific **Research Question (RQ)**. By systematically integrating relevant research works, SLR ensures a comprehensive review and analysis of relevant studies. Therefore, SLR is beneficial to establish fundamental knowledge for the chosen research topic.

In this article, SLR is utilized to explore the current research on DFL. Screening for relevant studies in each database involves carefully selecting work that fits the overall theme. This section outlines the steps for conducting an SLR in our work. First, identifying the research question is essential. Next, appropriate databases are selected for study search, keywords are screened, and rigorous selection criteria are established. The overall processes of our SLR flows are shown in Figure 3. The filtering process involves six academic database engines, yielding a total of 268 search results. Through additional screening mechanisms, irrelevant studies were excluded such as non-peer-reviewed, duplicate publications, survey papers, non-decentralized research methods, and articles not conducted on Non-IID data. Finally, as of the end of 2024, a total of 93 studies were retained after the screening process.

Table 2. Outline of Research Questions

Category	ID	Research Question	Motivation
Topic	RQ1	What are the current core challenges in these DFL and Non-IID studies?	To investigate the current DFL with Non-IID research and identify studies across different taxonomies.
Scenario	RQ2	What are the common application scenarios for DFL research nowadays?	Statistical analysis of various applications that can be leveraged in DFL.
	RQ3	Which datasets were used in the experiments?	Statistical analysis of the dataset that is being used in the current research.
Experiment	RQ4	What models were used for evaluating the experiments?	Every researcher has their own definition of a model, which leads to the ineffectiveness of aligning various research outcomes with quantifiable metrics.
	RQ5	What are the methods to implement Non-IID?	The widely accepted Non-IID definition leads researchers to have different implementations in their work.
	RQ6	Which performance metrics were used in the experiments?	Different scenarios have various ways to demonstrate method efficiency.

3.1 Formulation of Research Question

Based on the previous discussion, the lack of thorough investigation into challenges and issues associated with Non-IID data in DFL systems becomes evident. Therefore, this article aims to conduct an in-depth study and present an SLR of Non-IID data in DFL. Additionally, existing DFL experiments lack a unified performance evaluation metric. To address these issues, this article proposes six RQs, as outlined in Table 2. These six RQs are categorized into three types: taxonomy, scenario, and experiment. For the taxonomy, this article examines current studies and challenges concerning DFL and Non-IID data in Section 4. The scenario category mainly analyses the datasets discussed and used in the research works. Finally, this article integrates the selected research to elucidate how experiments were conducted, including the models used, methods for forming Non-IID data, and performance metrics for evaluating experiments. Detailed explanations of scenario and experiment categories are provided in Section 5.

3.2 Literature Search and Selection Methods

When using query keywords for searching, ensuring the selection of relevant and appropriate keywords is essential. Using the correct keywords with suitable logical operators (AND, OR) enables efficient narrowing down of the search scope, resulting in reliable research data. Therefore, this article first selected relevant keywords for RQ1, focusing on examining the impact of DFL and Non-IID. Due to the different abbreviation patterns used by authors when referring to Non-IID data, the keywords “non-IID” and “non-i.i.d.” have been included to ensure a more comprehensive survey. Accordingly, the final search terms are given as follows:

“decentralized federated learning” AND (“non-IID” OR “non-i.i.d.”)

While SLR aims to provide comprehensive research directions in the literature, this article conducted searches using query keywords in the following databases: (1) ACM, (2) IEEE Xplore, (3) MDPI, (4) ScienceDirect, (5) Springer, and (6) Web of Science. Although DFL first emerged in 2018, the search results indicate that the earliest published studies were found in 2021. Therefore, this SLR collects research papers related to Non-IID in DFL systems from January 2021 to December 2024. A total of 268 published studies are selected from the six mentioned database engines, including 63 studies from ACM, 29 studies from IEEE Xplore, 18 studies from MDPI, 69 studies from ScienceDirect, 74 studies from Springer, and 15 studies from Web of Science. However, the search results may include studies that are not related or irrelevant to the research. Relevant keywords

Table 3. Filter Rules in SLR

Filter	Rule	Description
1	Exclusion of Non-Peer-Reviewed Articles.	Exclude articles lacking a peer-review process. This selection criterion ensures that only peer-reviewed research is considered, as non-peer-reviewed preprints or books are deemed inadequate for quality assessment.
2	Exclusion of Duplicate Articles.	Exclude duplicate articles found across various databases. When a study appears in multiple databases, it is randomly chosen only once to avoid redundancy.
3	Exclusion of Survey, Review, and SLR Research.	Omit articles primarily comprising literature reviews without introducing novel methodologies. This criterion eliminates research that is solely focused on surveying existing literature, as such studies may not address the fundamental challenges in DFL.
4	Exclusion not focusing on DFL.	Eliminate studies that do not explicitly mention or discuss DFL within their content. This rule discards studies that merely reference DFL in their citations. Furthermore, the studies advocating for the continued use of a centralized server are filtered out at this stage.
5	Exclusion not focusing on Non-IID.	Discard studies that do not explicitly mention or discuss Non-IID. This criterion ensures that selected studies actively consider and address Non-IID aspects in their experimental design, eliminating those only reference Non-IID in their citations.

may only appear in the references and not be indexed. Therefore, manual filtering rules are established to ensure research quality. Table 3 outlines the rules for manual screening in this work.

To conduct an SLR of RQ1, this article classifies and summarizes selected research papers that address the negative impact of Non-IID data in DFL. The challenges are divided into four major taxonomies: architecture, topology, optimization, and security. Thereof, the architecture (Section 4.1) can be subdivided into two branches: partially decentralized and fully decentralization. The topology (Section 4.2) can be divided into similarity-based, classic-algorithm based, cluster, and performance metric. The optimization (Section 4.3) can be categorized into asynchronous, heterogeneous, and operating cost. The security (Section 4.4) is divided into blockchain, attack behavior, and trustiness. In summary, Section 4 organizes the selected research papers based on the above challenges and analyzes them in detail.

Section 5 presents the categories of Scenarios and Experiments, which address RQ2 to RQ6. At first, this article discusses the application scenario in Section 5. To identify the most common application scenarios in DFL studies involving Non-IID data experiments. Furthermore, this article describes the most common datasets used in experiments and how these relate to specific ML tasks. In the experimental part, our work explains common models, the typical Non-IID distribution methods, and the considered performance metrics. At last, this article investigates the most common performance metrics and whether the trends in these metrics vary over the years. Based on the above challenges, our work also proposes issues worthy of attention and potential research directions in the current DFL field in Section 6.

4 Taxonomy of Non-IID Problem in DFL

Decoupling various role components is necessary due to the decentralized nature. Compared with the CFL, clients in a decentralized architecture can simultaneously play one or multiple roles. This section briefly introduces the roles used in the DFL to support normal operations of the system. Typically, clients are categorized into three roles: training, aggregation, and committee. The relationships between these roles and clients are not limited to one-to-one bindings. Detailed descriptions for each subsection and corresponding branches are shown in Table 4 and Figure 4.

Table 4. Feature Descriptions and Insights for Current Issues in DFL

Section	Taxonomy	Description and Insights
Sec 4.1	Architecture	<i>Description:</i> Various architectures have been employed for FL training, ranging from partially to fully decentralized structures. <i>Insight:</i> Tradeoffs among convergence speed, communication cost, and robustness remain unresolved. Future research should explore adaptive hybrid architectures with lightweight fault-tolerant coordination to maintain scalability and robustness in dynamic large-scale systems.
Sec 4.2	Topology	<i>Description:</i> DFL employs diverse network topologies such as similarity-based, algorithm-based, and clustered structures to improve system performance, scalability, and fairness. <i>Insight:</i> No universal topology fits all DFL settings, indicating the importance of adaptive and performance-aware structures for balancing efficiency, stability, and fairness.
Sec 4.3	Optimizations	<i>Description:</i> Optimization in DFL focuses on improving communication efficiency, addressing heterogeneity, and minimizing operating costs. <i>Insight:</i> Challenges persist in addressing stale updates and tradeoffs between accuracy and cost, underscoring the need for adaptive optimization frameworks that ensure robustness and stability in Non-IID conditions
Sec 4.4	Security	<i>Description:</i> Encompasses blockchain integration, defense against attacks, and trust evaluation to ensure integrity, transparency, and confidentiality in DFL. <i>Insight:</i> Threats such as model inversion and unreliable nodes emphasize the importance of lightweight and adaptive security designs that maintain reliability without excessive overhead.

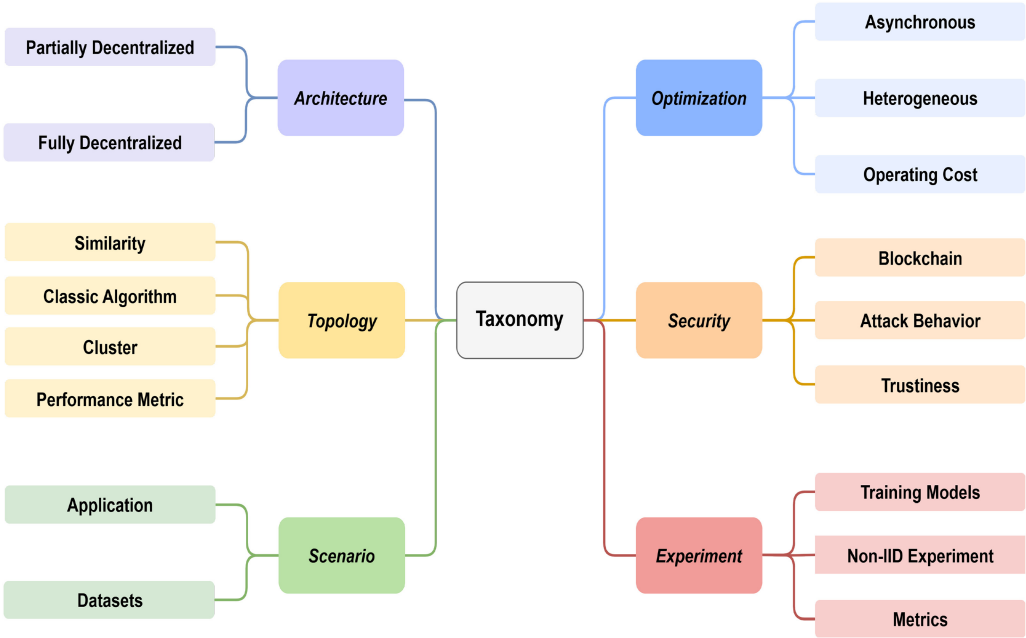


Fig. 4. Illustration of detailed topics and issues.

In the training task, each client assumes a fundamental training role. Upon receiving training instructions, clients engage in local training based on their privacy data to generate local models. Global models used by different methods may exhibit personalized characteristics because of the decentralized operating environment. The aggregation methods are diverse for the aggregation role due to the absence of guidance from a central server in the decentralized system structure. Common aggregation methods are broadly categorized into two types. First, the system selects one or more clients for aggregation. Second, each client possesses the identity for aggregation and

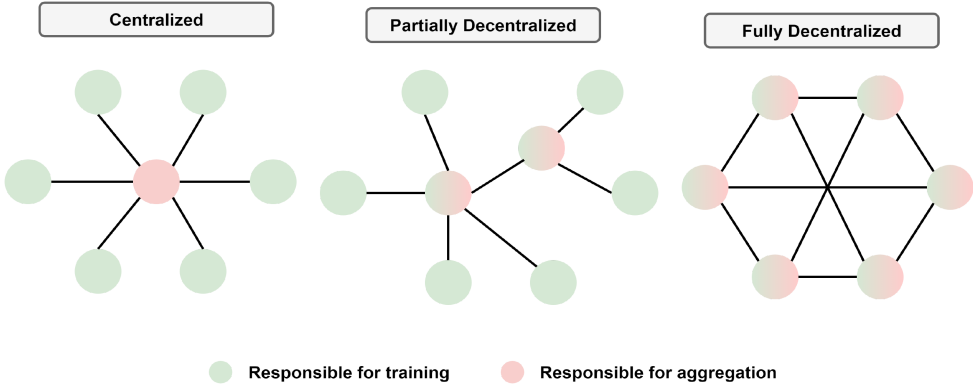


Fig. 5. Illustration of centralized, partially decentralized, and fully decentralized frameworks.

aggregates the received models from neighbors. Detailed discussions on aggregation structures are presented in Section 4.1.

Network topology issues have arisen in DFL. Section 4.2 discusses the topology of connections between clients. Finally, emerging research in DFL continues to explore existing issues in FL, such as communication costs and heterogeneity, which are thoroughly examined and addressed in Section 4.3. Moreover, the committee role serves as a common mechanism for dealing with security concerns. Considering the potential presence of malicious nodes in the system, additional mechanisms are introduced to select committees based on different indicators. Typically, committees are comprised of clients with higher accuracy or stable contributions. Section 4.4 conducts a detailed discussion on the committee election and security issues.

4.1 Architecture

The primary reason for transitioning from CFL to DFL is to resolve the single point of failure issue caused by a single-parameter server. However, various changes have occurred in the implementation of DFL research. Figure 5 showcases a shift from centralized to fully decentralized architectures. Solutions proposed for the relevant topic are categorized as Architecture Issues and composed of two types. First, the partially decentralized architecture involves selecting one or more clients through a consensus mechanism for aggregation tasks in a decentralized system. Second, the fully decentralized architecture allows each client in the system to possess the capability for training and aggregation, so as to avoid aggregation failures.

4.1.1 Partially Decentralized Architecture. In DFL, the server role responsible for aggregation in CFL no longer exists. How to take the role of aggregating models during the training process is a challenge. Current research papers have proposed several classification criteria to select one or more clients responsible for aggregation from the system. These criteria include evaluations based on contributions, latency, or honest scores based on computational nodes. This article categorizes such studies under the branch of partial decentralization.

To select the aggregator in the DFL training process, the study [29] adopts contribution scores as the selection criterion and employs knowledge distillation [30] to choose nodes with the highest contribution scores for aggregation work. In LFNL [31], a leader election algorithm based on latency is introduced to select the role of a central node in edge devices, so as to reduce overall latency. A committee mechanism is employed in CE-Fed [32]. Various leader nodes in the group are allowed to enter the model aggregation committee to facilitate the aggregation of the global model.

Some studies have also used a random selection of clients as aggregators. In FRACTAL [33], clients are first grouped based on communication delay, computing power, and data distribution. Then, one client is randomly selected from each group as the aggregator for this group. These group aggregators form new groups, and a new aggregator is selected for further aggregation, gradually creating a multi-level topology. BlockDFL [34] maps clients to hash rings using the hash value from the previous block. The size of allocated segments is proportional to the shares of clients. Clients with more shares are more likely to be selected as aggregators or validators, while other clients are update providers. Roles are redistributed at the start of each round to ensure fairness and prevent bribery attacks.

Another study [35] proposes an improved training strategy for BC-DFL to select clients based on the class imbalance reduction algorithm [36]. The selected client constructs a dataset through network construction to reduce bias. Meanwhile, the unselected clients act as miners and aggregate local models into a global model. Furthermore, honest scores can also serve as a criterion for selecting clients. ABFL [37] calculates the honest scores of nodes, where a higher honest score indicates a node beneficial to model training. Using a hash function to select proxy nodes, nodes with higher scores have a greater probability of being selected. After selecting a proxy node, a super-proxy node is then chosen from the proxy nodes to aggregate models from all nodes.

Finally, the selection mechanism for multiple parameter servers explores scenarios based on factors such as training time, device connectivity, and energy consumption [38]. The comprehensive consideration contributes to effectively utilizing system resources and making the system more equitable.

4.1.2 Fully Decentralized Architecture. Some studies propose methods to achieve a fully decentralized architecture during the training process. This approach enables all clients in the system to possess aggregation capabilities rather than limiting the aggregator role to specific clients. Each client exchanges training models from designated neighboring nodes in this architecture. Aggregation is performed based on different methods to produce generalized or personalized models. RDFL [39] and RDFL-CS [174] adopts a structured network with virtual positions for all clients. Each client locates neighbors based on positions within a ring. The AllReduce mechanism [40, 41] is applied to acquiring more parameters from neighbor models gradually while reducing operational costs.

To enable each client to adapt to model aggregation dynamically, the study [42] explores exchanging local models with another learning pair, enhancing system fairness. In ProxyFL [43] and FedSCD [44], each client maintains a public proxy model and a private model. ProxyFL uses **Stochastic Gradient Push (SGP)** [45] to allow clients to receive specified neighbor models, aggregate the public proxy model in each round, and then update the private model based on gradient deviations. The heterogeneous model structure helps prevent model attacks and reduces network traffic during transmission. On the other hand, FedSCD introduces soft label transfer through knowledge distillation, allowing private models of less influential clients to learn more from the proxy model. As a result, training becomes more efficient, and privacy protection is enhanced.

In the study [46], each client, while collecting neighbor models, tests the accuracy of their local model on a collectively owned small dataset. The test accuracy is then used as the basis for clients to determine the proportionate weight of neighbor models during the aggregation phase. Another study [47] employs GANs to generate a blended dataset for the private dataset of each client. Clients can transmit the training model and the blended dataset to neighbors. Receiving nodes adjust the incoming model using their own dataset and the received blended data to avoid catastrophic forgetting issues. These methods contribute to enhancing model performance and system efficiency.

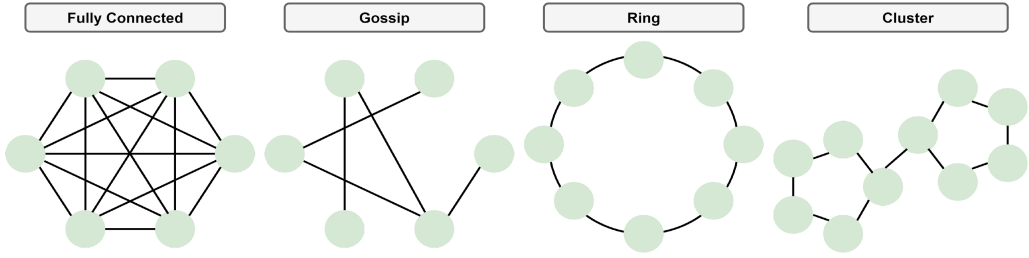


Fig. 6. Illustration of network topologies.

For distributed fault diagnosis using **federated domain generalization (FDG)** methods, FDG-CA [48] improves the existing FDG approach. The method incorporates the concept of DFL, which allows each client to independently exchange model information. In the healthcare domain, model privacy and effective integration are among the most critical challenges. To address these issues, AccDFL [175] introduces a DFL architecture into **Healthcare IoT (HIoT)**. The approach effectively tackles the problems of slow convergence and communication bottlenecks commonly encountered in traditional medical applications. The efficiency of decentralized collaborative learning in large-scale medical IoT networks is also significantly enhanced.

4.1.3 Comprehensive Discussions. Section 4.1 explains the evolution from CFL to DFL. DFL introduces two main types: partially and fully decentralized to address the single point of failure issue in CFL. DFL architecture aims to improve system robustness, scalability, and fairness.

Partially decentralized architectures assign aggregation roles to selected clients in each training round based on contribution scores, latency, trust scores, or randomness. Methods including knowledge distillation, delayed election, committee-based selection, random grouping, and hash ring mapping are commonly adopted to enhance fairness, reduce vulnerability to attacks, and support Non-IID data and device heterogeneity. Additional factors such as training duration, device connectivity, and energy efficiency further improve resource utilization and system flexibility. In contrast, fully decentralized architectures allow all clients to participate equally in model aggregation through peer-to-peer communication. Techniques such as ring topology, AllReduce operations, random pairing, proxy and private model separation, soft-label knowledge distillation, and GAN-based synthetic datasets support robust control without central coordination while enhancing privacy protection and adaptability to diverse system configurations.

Partially decentralized architecture balances aggregation role assignment and coordination efficiency, making the design suitable for semi-trusted environments. Fully decentralized architectures provide stronger resilience and privacy guarantees, and are more applicable to large-scale or highly heterogeneous systems, such as medical IoT. Tradeoffs between convergence speed, communication overhead, and system robustness should guide the architectural choice.

4.2 Topology

The DFL network topology comprises diverse classic P2P structures, such as fully connected graphs, gossip graphs, ring graphs, cluster graphs, and so on. Figure 6 provides a detailed illustration of the classic communication architecture in DFL. The classification of these architectures is based on a comprehensive review of SLR filtering research papers. Various criteria are considered to determine network topology, encompassing factors such as similarity, classical algorithms, clustering, and performance metrics. These indicators aid the client in constructing the network topology.

4.2.1 Similarity. The similarity can be further divided into two main themes: data similarity and model similarity. First, data similarity relies on the distribution of data categories among clients. Accordingly, DFL-DF [49] and Fed-SHARC [50] employ cosine similarity calculation based on data features of a node to determine the probability of selection. DFL-DF chooses nodes with different data distributions as neighbors to form a more generalizable global model. On the other hand, RDFL-CS [174] determines the position of each node in a ring topology based on the cosine similarity of data features between nodes.

Next, addressing model similarity is one of the most used reference indicators. In DACFL [51], a dynamic average consistency algorithm [52] is used to track the models of neighboring nodes and determine the network topology, transforming the aggregation of FL models into a dynamic average consistency problem. The goal of the study [42] is to enable clients to find another learner to exchange local models, ensure fairness among all clients, and avoid subsets of clients gaining greater weight. In CoCo [53] and RCEDL [54], the researchers employ Euclidean distance and cosine similarity distance to measure the differences between each local model and the average of other local models. The system selects nodes based on the size of consensus distance to form a topological structure. Meanwhile, considering communication capacity, the topological structure can dynamically adjust for more efficient model aggregation.

In AsyNG [55], the study relies on a priority algorithm to select neighboring nodes based on loss value, communication frequency, and local update ratio. Through a greedy approach, AsyNG chooses a subset of neighboring nodes to form a consistent model as much as possible. Finally, the research in NILM [56] clusters clients based on the power consumption and computational resources of devices in IoT applications. Clients are connected with similar computation costs to form multiple levels of clusters.

4.2.2 Class Algorithm. Due to the decentralized nature of DFL, many studies in DFL leverage various classic P2P algorithms to make corresponding improvements. Building upon existing P2P algorithms, these research studies propose various communication topology architectures.

For instance, RDFL [39] incorporates the Ring Chord algorithm [57] into the AllReduce method. By employing a circular topology, RDFL enhances reliability and performance within the DFL framework. The construction of a Ring Chord is similarly applied in the studies [58, 59]. GossipFL [60] and DFL-ATC [176] introduces the Gossip Protocol [61] for each client to communicate with only one peer. The peer is dynamically selected through the Gossip Matrix for more efficient communication budget utilization. The one-to-one communication method considers choosing peers with minimal bandwidth, maximizing bandwidth utilization, and improving communication efficiency.

DPG [62] enables each client to construct an L-hop neighbor list. Communication nodes are selected based on probability vectors, considering similarity and communication cost. The approach allows for the accurate training of personalized models at acceptable communication costs. Clients with higher selection probabilities are more likely to become communication targets and request models from neighboring nodes via broadcasting. In ProxyFL [43], clients select different neighbors in each round based on the SGP [45] through an exponential communication protocol. Therefore, network traffic is reduced significantly in large-scale systems. TOPFL [63] uses a greedy algorithm to dynamically adjust P2P topology based on available communication and computing resources of the client. Remove slow connections from the original training topology to alleviate communication overhead.

By integrating and enhancing algorithms, various communication topology architectures are proposed to improve the performance and efficiency of DFL systems.

4.2.3 Cluster. In addition to treating each client as an independent entity, the technique of clusters or groups has been developed. In this context, clusters become the fundamental units for overall training. Each cluster potentially comprises multiple clients. Clustering conditions may be based on data categories and may even be related to the training topology structure. The following subsection delves into studies related to network connections.

H-DFL [64] employs MergeCBF [65] to represent data distribution [66] compactly. Clusters with complementary data are gradually merged before training begins to form multiple fixed clusters. Pre-grouping based on data distribution statistics maximally alleviates data imbalance issues within each cluster. In TH-DFL [67], a hierarchical structure is used to group clients into clusters. Each cluster includes executing clients and supervisory clients. After local training, executing clients send encrypted local gradients to supervisory clients. Supervisory clients calculate the average gradient and send the results back to executing clients for performing aggregation. Delay is also used as a classification indicator. CE-Fed [32] compares the latency levels among clients and enables clients with similar latency levels to gather in the same group.

These concepts of clusters or grouping introduce more organizational and collaborative elements into DFL systems. Therefore, data distribution imbalances and system performance can be improved.

4.2.4 Performance Metric. In the study of performance evaluation metrics, some studies focus on the client contribution as the basis for selecting neighbors to construct the final communication topology. The study [29] employs the knowledge distillation method to assess the contribution scores of neighbor models. The selection of neighbors is determined to optimize the overall communication efficiency of the system.

In addition to the research on fixed topologies, the study [68, 177] has explored communication topologies with time-varying updates. In this scenario, the interconnection topology among clients changes dynamically while all clients remain constant. When the communication topology undergoes alterations, the study utilizes the Metropolis-Hastings method [69] to update the weighted adjacency matrix based on the current communication topology. Therefore, the convergence of average consensus can be ensured. Immediate notifications are sent to connected neighbors when clients leave or join the training. The system reflects the status of training clients before the next training round.

These two aspects of evaluations have collectively enriched the understanding of performance metrics. Researchers consider both the client contribution levels and adaptively respond to dynamic topological changes. As a result, the efficiency of distributed learning systems can be enhanced.

4.2.5 Comprehensive Discussions. Section 4.2 explains the design and types of network topology in DFL. The discussion includes classic P2P structures and dynamic communication architectures. Network topology directly impacts model aggregation efficiency and also plays a critical role in determining system scalability, fairness, and fault tolerance. Four key aspects are discussed: similarity, classic algorithms, clustering, and performance metrics.

Some topologies exploit data or model similarity to connect clients. Data similarity uses local dataset distributions and cosine similarity to select neighbors, supporting generalization and structured placements like ring structures. Model similarity compares parameters using Euclidean or cosine distance to ensure fair and efficient aggregation. Some designs also group clients by loss, communication frequency, or computing power. These groupings create hierarchical structures that reflect the characteristics of clients. Many DFL approaches adapt P2P algorithms such as Ring, Chord, Gossip Protocol, and All-Reduce. These structures improve reliability, reduce communication load, and enhance bandwidth usage. Techniques like L-hop neighbor selection and

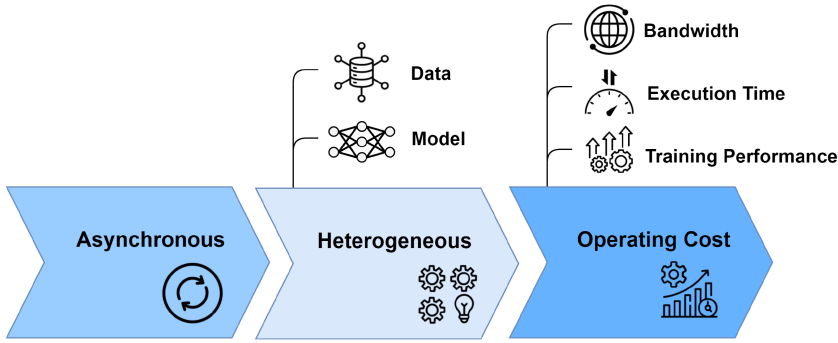


Fig. 7. Illustration of optimization issues.

broadcast can balance cost and accuracy. Clustering organizes clients into groups based on data distribution, latency, or computing resources, with each cluster acting as a training and communication unit. Pre-grouping mitigates Non-IID effects and enhances local aggregation, while hierarchical structures with supervisory roles improve coordination and collaboration. Performance further improves through dynamic topology updates, client contribution-based neighbor selection, and methods like Metropolis-Hastings to maintain stability and convergence in changing environments.

These observations suggest that no single topology design universally outperforms others across all DFL scenarios. Similarity-based structures improve fairness and generalization but may increase computational cost. Algorithm-based designs such as ring and gossip enhance communication efficiency. Clustering addresses data heterogeneity and strengthens local coordination, while performance-aware topologies adapt dynamically to optimize convergence and stability. Each approach reflects a tradeoff among efficiency, accuracy, and adaptability in decentralized learning systems. The effectiveness of a given topology often depends on factors such as data heterogeneity, network conditions, and task specific requirements.

4.3 Optimization

In the optimization section, as depicted in Figure 7, this article systematically organizes and categorizes the research into three major classes. The first category is the asynchronous aspect. Due to the long-standing challenge of accommodating slow clients in FL, decentralized frameworks have inadequately addressed the issue of the overall training imbalance. Therefore, this article isolates the asynchronous problem as an independent section for in-depth exploration. The second category pertains to heterogeneity, including data heterogeneity and model heterogeneity. The last category is encompassing aspects related to bandwidth, execution time, and training performance.

4.3.1 Asynchronous. Asynchronous training in FL has been a topic worthy of in-depth exploration. Numerous studies have focused on various aspects of asynchronous training in CFL. Similarly, asynchronous training remains a relatively unexplored area in DFL. Hence, this section delves into studies related to asynchronous aspects of DFL.

The AsyNG framework [55] and the AsyDFL framework [178] have introduced an asynchronous training framework for DFL. AsyNG employs directed gradient pushing to allow clients to send parameters to neighbors without waiting for a response. FedLyra [70] utilizes a flag to flip and stop collecting models for the current training round before waiting for a certain proportion of local models. Asynchronous execution reduces the risk of deadlock and enhances training efficiency, particularly addressing the issue of low efficiency caused by heterogeneous devices. DeFTA [71]

achieves synchronization in the DFL architecture by synchronously maintaining multiple sub-FL systems.

The study [72] proposes a novel blockchain architecture that leverages blockchain-preserving contribution proofs of ML models. The proposed method adopts an asynchronous gossip training method to store proof of contribution. Each node is allowed to asynchronously generate its own block and self-maintain, so as to enhance overall training efficiency. Reputation is earned through the calculation of accuracy and contribution proofs, concurrently preventing malicious node attacks. DHFDL [73] combines blockchain and knowledge distillation to obtain personalized models through loop knowledge distillation technology. Then upload the trained model to the blockchain for other clients on the chain to download the model for local training.

4.3.2 Heterogeneous. The data heterogeneity in FL focuses on the Non-IID data issue. In CFL, prior work has investigated methods to mitigate data heterogeneity using techniques such as client selection, knowledge distillation, and adaptive aggregation [186]. However, DFL lacks a central coordinator, which complicates client selection and removes the global view needed for adaptive aggregation. To address these issues, this article categorizes relevant studies and conducts an in-depth analysis to reveal the challenges and coping methods when dealing with Non-IID data in the DFL environment.

(a) Data

For heterogeneous data, numerous studies transform client datasets in various ways. For instance, YOGA [74] utilizes a coordinator to gather client bandwidth and data distribution. YOGA ranks peers based on learning speed and hierarchical differences. The higher priority is assigned to peers with greater available bandwidth and data distribution differences. H-DFL [64] adopts mergeable Bloom Filters [65] to efficiently represent the local data distribution of each client. Marking data distribution through Bloom Filters facilitates comparison of datasets between different nodes. Def-KT [75] applies the **Mutual Knowledge Transfer (MKT)** strategy [76] in DFL, effectively addressing data distribution differences among different clients. DFedSN [77] introduces the concept of Affine Distribution Shifts [78], allowing data in clients to affine into disturbances, which are incorporated into the model gradients to enhance robustness.

On the other hand, research has been conducted to optimize model efficiency for heterogeneous data. The introduction of a **Directed Acyclic Graph (DAG)** structure, as proposed in the Specializing DAG [79], treats the model updating process as a DAG. Specializing DAG enhances the randomness of model updates through a stochastic selection based on accuracy perception. NET-FLEET [80] incorporates a recursive gradient correction technique into the local update scheme in FL, achieving linear acceleration in convergence speed. Improved specialization and generalization capabilities under Non-IID distributions are realized through cluster structures and indicator metrics.

Knowledge distillation, widely applied in research [81–84, 179, 180], guides student models using the predictive results of teacher models. A common transfer set is employed for implementation. DAITA [83] utilizes multiple teacher models in knowledge distillation, leading to enhanced model performance. In the study [82], an adaptive ensemble learning framework is employed for dataset classification. Grouping data of the same category together helps mitigate the impact of Non-IID data. Another study [179] mitigates data distribution differences across devices through transfer knowledge integration. In order to quickly and effectively train the global model under Non-IID data, Tram-FL [85] proposed a dynamic model routing algorithm. Prioritize clients with less specific label data to prevent the model from being too biased towards certain labels, thereby improving the overall performance and fairness of the model.

(b) *Model*

Due to infrastructure disparities among client devices, the overall training process is affected. In addition to data heterogeneity, some research papers further delve into model heterogeneity among different clients.

For heterogeneous models, knowledge distillation guides the student model to learn from the predictive results of the teacher model, eliminating the need for homogeneity among all client models. KD-PDFL [86] integrates knowledge distillation, employing Wasserstein distance during node-wise training of neighboring models to calculate differences and ensure that the models undergo moderate changes. FedLCon [87], through the mechanism of consensus rounds, updates after specific iterations, ensuring model parameter consistency through multiple communications.

MCFL [88] divides clients into multiple clusters. Each cluster trains its own model and learns different knowledge between clusters through transfer knowledge. The method uses a model difference tolerance formula to select client candidates for aggregation. Large discrepancies are avoided during the process, effectively filtering out updates with significant differences from local updates. In SLFed [89], clients exchange their GAN datasets and use subjective logic [90] to obtain similarity, dissimilarity, and uncertainty scores. The process helps select suitable clients for model aggregation and allows each client to build their own FL model flexibly.

4.3.3 Operating Cost. In the field of FL, numerous existing research topics still require further in-depth exploration. Simultaneously, in the domain of DFL, there is substantial discourse on operational costs, encompassing issues such as bandwidth, execution time, and training performance. Researchers pose numerous challenging questions regarding the future development of FL, demanding deeper investigation and analysis for resolution. Concurrently, as a distinctive branch of FL, DFL primarily focuses on effectively reducing operational costs. This includes enhancing communication efficiency, reducing training time, and optimizing training performance.

(a) *Bandwidth*

One of the issues is to discuss the utilization of bandwidth resources during the training and transmission process. A straightforward approach to reduce bandwidth is by decreasing the number of connections in transmission. In RDFL [39], FedISP [58], and CSL [59], AllReduce is integrated into DFL, where each client only transmits its local model to the next neighbor, significantly reducing network overhead and addressing network congestion issues. In ProxyFL [91], clients employ SGP [45] to select a neighbor in each round and additionally maintain a smaller model responsible for external transmission. ProxyFL proves effective in significantly reducing network traffic during transmission in large-scale systems. CE-Fed [32] introduces a clustering mechanism through an aggregation committee in Secure **Multiparty Computation (MPC)** to minimize bandwidth usage.

Additionally, many studies focus on reducing bandwidth through model size. CoCo [53], RCEDL [54] and BlockDFL [34] utilize model compression to decrease communication during model training. However, CoCo considers the heterogeneity between clients, which allows nodes to independently set compression ratios to avoid transmission delays. GossipFL [60] performs sparsification on peer client model parameters before exchanging, significantly lowering communication costs. In DFL-DF [49], each client model includes a feature extractor for data feature extraction and a classifier for prediction. Transmitting data features instead of the entire model helps reduce communication overhead. YOGA [74] employs a hierarchical model aggregation strategy to effectively share information among multiple clients, aiming to enhance overall model performance. Further reduction in bandwidth is achieved by improving the communication approach of GossipFL [92] through hierarchical model splitting, quantized transmission to lower

node-to-node communication costs. Simultaneously, an adaptive learning rate adjustment is proposed to accelerate overall training effectiveness by adjusting the learning rate based on training round numbers and the extent of loss reduction to enhance training accuracy.

To effectively improve overall training efficiency and reduce communication resource consumption, ST-LF [93] assesses data quality by evaluating the model trained on each device. Devices are classified based on data quality, and model transmission links between clients are optimized. TOPFL [63] splits the model into a personalized submodel and a base submodel. Clients retain the personalized submodel and only share the base submodel with neighbors. Transmitting only part of the model parameters instead of the entire model reduces transmission costs. DTFed [94] adopts the concept of knowledge distillation, extracting knowledge from local models and using soft labels to replace the traditional method of transmitting model parameters. DFLStar [181] performs parameter exchange based on the similarity of the final model layer to minimize communication overhead. A strategy is also adopted in which only one neighbor is selected for model aggregation in each communication round, significantly reducing cross-network transmission traffic.

In blockchain-related research [39, 95–99], the use of the **InterPlanetary File System (IPFS)** [100] as a decentralized file storage system for storing models from different clients is explored. IPFS also contributes to lowering communication GAS [101] costs within the blockchain.

(b) *Execution Time*

In the BLADE-FL [102] framework integrates blockchain with FL, each client simultaneously conducts training and mining in a round. Consequently, the time spent in each round is longer compared with traditional FL. The study [35] divides the nodes on the blockchain into clients responsible for training and miners responsible for mining, aggregating models, and generating new blocks. Client selection ensures that only clients contributing to improving model accuracy participate in training, thus reducing performance degradation. Additionally, synchronous training and mining help decrease the overall training time.

TSS SGD [103] introduces different parameter transmission methods based on topology. Devices with multiple connections play the role of “hubs.” The topology structure among network devices determines whether a device should perform swap or skip operations after receiving parameters from other nodes. Swap operation is chosen when directly connected to the device, enabling the transmission of its own parameters. Skip operation is selected when the device is indirectly connected through other devices, requiring the passing of parameters from the previous node. Redundant model transfer process is reduced. The study [104] accelerates training convergence by adjusting the number of P2P connections. A limited number of k edges are introduced in the P2P graph to facilitate DFL convergence. The analysis explores the relationship between model training convergence rate and the second smallest eigenvalue of the graph’s Laplacian matrix. The most advantageous edges are selected to maximize this eigenvalue and thus accelerate the convergence speed.

(c) *Training Performance*

In the study [47], the received datasets from neighboring nodes are retained in the buffer for subsequent use. Training the model using the datasets in the buffer helps retain previously acquired knowledge to prevent catastrophic forgetting issues. Regarding the aggregation approach, FedISP [58] introduces incremental subgradient-proximal methods [105] instead of direct aggregation, ensuring model consensus among clients for better convergence.

Due to the respective drawbacks of CFL with a central server and fully DFL, the study [106] proposes a semi-DFL framework with a two-layer structure. Device-nodes are grouped into **Device-to-Device (D2D)** clusters based on distance. Within D2D clusters, local aggregation takes place,

and randomly selected device nodes upload their models to the central server for global aggregation. Singular value metrics are calculated based on node adjacency matrix to understand the connectivity structure of D2D clusters. A dynamically changing threshold is designed over time to influence the number of client nodes required for the central server aggregation. Improved convergence speed while reducing **Device-to-Server (D2S)** communication.

In past DFL frameworks, all clients had the same training strategy, preventing effective utilization of resources for clients with different computational or transmission capabilities. Therefore, the study [107] proposes the **Incentive-aware Decentralized Federated Learning (IDEA)** framework, allowing customization of incentive functions to encourage participation and data contribution. It introduces a **Multi-Agent Reinforcement Learning (MARL)** approach [108], enabling each client to choose its optimal collaboration strategy.

4.3.4 Comprehensive Discussions. Section 4.3 discusses optimization problems in DFL. The discussion focuses on asynchronous training, heterogeneity handling, and operating cost. These challenges arise from the decentralized nature of the system. Client devices often differ in data distribution, computational capabilities, and communication resources.

Asynchronous training reduces the straggler problem by allowing clients to send updates independently. Some systems start aggregation based on trigger conditions, and others use blockchain to record contributions and enhance security. Data heterogeneity is addressed through client grouping based on bandwidth or data distribution, and through techniques like Bloom filter-based distribution comparison and affine transformations that improve robustness. Model heterogeneity is managed through knowledge distillation between different model structures and synchronization using consensus rounds. Operating cost is reduced through bandwidth-saving techniques such as model compression, sparsification, and partial model transmission. Memory reuse, incremental updates, and selective collaboration further improve training efficiency.

Moreover, asynchronous training improves system responsiveness by allowing flexible update schedules. However, significant variation in client training speeds may cause stale updates and lead to unstable convergence. Heterogeneity-aware methods improve robustness under Non-IID conditions. Grouping strategies and distillation parameters require careful tuning to minimize communication overhead and maintain training effectiveness. Cost-saving techniques such as model compression and sparse transmission reduce bandwidth usage. Excessive reduction of model size or transmission frequency may lead to degraded model accuracy. The effectiveness of each method depends on factors such as client hardware variability, data distribution imbalance, and network reliability. Adaptive integration of optimization strategies is essential for maintaining training stability and ensuring consistent performance in decentralized environments.

4.4 Security

While the core concept of FL involves keeping the client raw data locally, there are still risks. Specifically, a risk exists in inferring the types of data the client possesses from transmitted model parameters and other data. For this reason, numerous research papers have explored privacy and security issues under the FL framework. In this survey, security and privacy concerns in DFL are addressed. The incorporation of technologies such as blockchain, differential privacy, and committee systems within the DFL framework is discussed in the Security section. Refer to Figure 8 for specific details.

4.4.1 Blockchain. Blockchain technology [109] inherently possesses the characteristics of a decentralized system, while also incorporating functionalities such as reliability, immutability, transparency, and security. Consequently, numerous studies have integrated FL with blockchain technology, forming a decentralized system framework. Many studies [81, 99, 110–114] are based

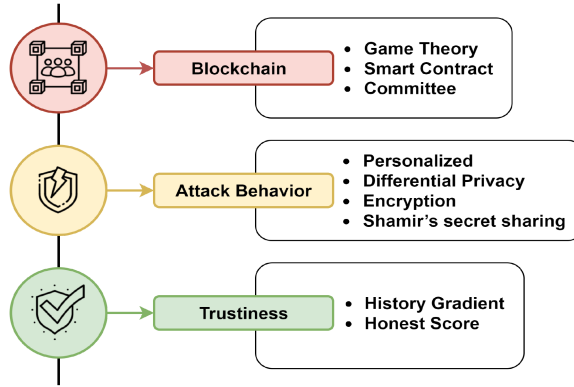


Fig. 8. Illustration of security issues.

on research involving blockchain. Blockchain is a decentralized data storage and transaction system that utilizes distributed nodes to verify and record transactions. According to the above characteristics of blockchain, applications built on this technology can maintain decentralization and security.

Differentially Private Stochastic Gradient Descent (DP-SGD) [115] has been applied in numerous existing studies [111, 116], incorporating noise into the gradient descent process to safeguard the models transmitted by each client. To incentivize active and rational participation in training, the study [117] introduces Rational Trust Modeling [118]. The application of game theory methods on blockchain aims to improve the quality of training. PoShapley-BCFL [97] and study [119] employ Shapley values to assess the contribution of each client gradient descent to the model and employ these contributions to determine weights during aggregation.

Due to potential issues in traditional FL like a single point of failure, malicious nodes, or a lack of incentive mechanisms, CABFL [98] proposes an FL approach that combines blockchain technology. CABFL employs smart contracts to secure model selection and training processes. Through smart contracts, clients can select a specified number of models from the recently submitted model list. After model testing, the most suitable model for client dataset is chosen for aggregation based on accuracy. The constraints of smart contracts effectively prevent malicious nodes from submitting harmful models. Simultaneously, a reward mechanism encourages clients with outstanding computational capabilities, thereby enhancing overall training effectiveness and speed.

Addressing Byzantine attacks, research [39] and PoS-BFL [120] employ committee election methods [121] to select nodes with poor accuracy performance on the blockchain, reducing issues caused by malicious checkpoints. FedLyra [70] utilizes a reputation mechanism and establishes a **Practical Byzantine Fault Tolerance (PBFT)** committee. ComAvg [122] allocates clients to different committees and enhances committee client ability to resist Byzantine attacks through Krum [123] and Median [124] scoring mechanisms.

4.4.2 Attack Behavior. In the domain of deep learning, various attack techniques are known. DEFEAT [125] exhibits resistance against gradient attacks, allowing each client to communicate solely within a one-hop neighborhood. Provide each client with an independent, personalized model to prevent global model vulnerabilities from being exploited. In ProxyFL [91], clients are required to maintain both a public proxy model and a private model, creating a heterogeneous relationship. By transmitting only the differentially private public proxy model, ProxyFL mitigates

model attacks [126, 127]. Differential privacy is also employed in sensitive data scenarios, such as healthcare [110], **Smart Grid (SG)** [128], and Multi-Agent Systems [129].

Fed-SHARC [50] uses differential privacy for model aggregation within clusters by adding random noise to protect client data privacy. Model aggregation between clusters is performed using MPC, enabling multiple clusters to collaboratively compute the aggregation results without disclosing each other's data. DP-DZOGT [128] develops a **One-Point Zeroth-Order Gradient Estimator (OPZOG)** to predict the gradient direction. Differential privacy noise is incorporated to safeguard the privacy of transmitted information while accelerating convergence. Another study [129] combines differential privacy with bandit gradient estimation in **Push-Sum Distributed Dual Averaging (PS-DDA)**, using differential privacy to protect privacy and compute true gradient decisions to guide client updates.

Concerning Gradient Inversion Attacks, DFL-G [130] demonstrates effectiveness in countering attacks in a decentralized environment. Their work makes it impossible for attackers to easily extract raw materials from the DFL global trust model. In the study [96], adversarial behavior issues are initially addressed, utilizing the Pedersen commitment scheme [131] in blockchain to enhance the security of model transmissions. MChain-SFFL [132] adopts a multi-chain approach, using unique random numbers in each chain to mask the weights of clients, ensuring privacy without compromising accuracy. CE-Fed [32] employs secure MPC, allowing client models to be transmitted in an encrypted form within a P2P framework and then aggregated. The 2DF-IDS [116] incorporates quantum algorithms into the transmission model, employing a more secure communication protocol for connectivity.

Considering the potential vulnerability of model inversion attacks during the transmission of models between clients, the study [68] proposes the integration of **Shamir's Secret Sharing (SSS)** algorithm [133]. Leveraging the core concept of SSS, the local model is distributed to adjacent nodes. After receiving partial models from neighbors, each client updates models by employing a consensus algorithm to sum up the owned partial models.

4.4.3 Trustiness. In the context of research on other protective models, credibility issues are also explored. In DeFTA [71], the assumption is that all clients are selfish. Each client continuously evaluates the models sent by all peer nodes to decide whether to disconnect peer nodes considered faulty. DFedHG [134] applies historical gradients [135] in DFL to distinguish between normal, untrusted, and malicious clients. DFedHG further explains how to effectively utilize data from untrusted clients. TH-DFL [67] introduces a Trustiness-based mechanism in the DFL framework. The framework aims to strike a balance between ensuring privacy and model robustness while addressing challenges in defending against Byzantine attacks.

The study [182] designs an evaluation mechanism based on a "dual reputation score", which combines model cosine similarity and local validation loss to enable each node to dynamically assess the trustworthiness of the neighbors' models. Neighbor selection and aggregation strategies are then adjusted accordingly. DFL-ATC [176] quantifies the historical performance of neighboring nodes and employs a reward-punishment system to dynamically adjust trust scores, thereby enhancing stability in heterogeneous data environments. BALANCE [183] evaluates trust based on the similarity between local and neighbor models. When a neighbor model significantly deviates from the local model, the node is considered potentially malicious and subsequently ignored.

BlockFlow [95] adopts a reward distribution mechanism in the blockchain to ensure the system operation and incentivize clients to remain honest. Core mechanisms include a quality assessment model to ensure clients provide truthful and high-quality contributions. Assessing models through quality helps prevent malicious node attacks that could compromise the effectiveness of model training. ABFL [37], based on the helpfulness of nodes in model training, calculates honest scores

to predict the future direction of model updates. ABFL compares proxy nodes with high honest scores to the general model and predictive model. This filtering process aims to identify potential malicious nodes, reducing their impact on the system. PPT [184] uses a symmetric encryption system to ensure secure communication between network nodes.

4.4.4 Comprehensive Discussions. Section 4.4 explores the design of security and trust mechanisms in DFL, covering three key areas: blockchain integration, defense against attack behaviors, and trust evaluation. These components address vulnerabilities in decentralized environments, such as data exposure, malicious behavior, and unstable collaboration. Ensuring confidentiality and trust is essential for deploying DFL in sensitive or high-risk scenarios.

Some studies employ blockchain to ensure data integrity, decentralization, and transparency across distributed participants. Smart contracts are used to automate model selection, aggregation, and reward distribution without requiring a central server. Reputation and contribution scoring systems help quantify node reliability, which is essential in P2P environments. Differential privacy is commonly adopted to protect gradient and model information during communication, especially in sensitive domains such as healthcare. To mitigate risks such as model inversion and gradient leakage, various frameworks introduce proxy models, secret sharing, and encrypted transmission. Trust mechanisms are often built on historical behavior, model performance, or similarity metrics to evaluate neighbor credibility. Several designs further incorporate reward and punishment strategies and dynamic trust weighting to promote honest participation and ensure stable aggregation in heterogeneous or adversarial conditions.

Although DFL inherently enhances data privacy by keeping data local and removing centralized servers, security risks still persist. Transmitted model updates can be exploited through attacks such as gradient leakage or model inversion, and unreliable or malicious nodes may compromise model integrity. As a result, incorporating additional security mechanisms remains essential. Techniques like blockchain ensure transparency and immutability in coordination processes, while differential privacy helps prevent exposure of sensitive data during communication. Proxy models, secret sharing, and encryption further strengthen protection against adversarial behavior. These mechanisms are fundamental in highly sensitive information applications like healthcare, finance, and critical infrastructure. Protecting sensitive data, ensuring reliable system behavior, and maintaining secure collaboration justify the added complexity and resource cost of security.

5 Applications and EXperiments

This section addresses the RQ2 through RQ6 to this SLR, particularly in delving into the practical applications of DFL, the selection of related datasets, the utilized neural network models, the implementation methods for Non-IID data experiments, and the performance metrics discussed in the filtered studies. This article presents data and trends from filtered studies and illustrates pie charts for clarity.

5.1 Application Scenarios

To address RQ2 on common DFL application scenarios, Figure 9 classifies the applications discussed in the filtered studies. Figure 9 shows the variety of DFL applications and the proportion that elaborates on its application contexts. In the 93 filtered studies collected by this article, only 30 studies mention their application contexts. More than half of the studies do not mention their application scenarios, while some studies mention more than one application ($N=35$). IoT and Healthcare account for 29% and 26% of the applications, respectively. Transportation, including the application of **Unmanned Aerial Vehicles (UAV)**, self-driving cars, and the **Internet of Vehicle (IoV)**, follows with 11%. Edge computing stands at 11%, while intelligentization, which

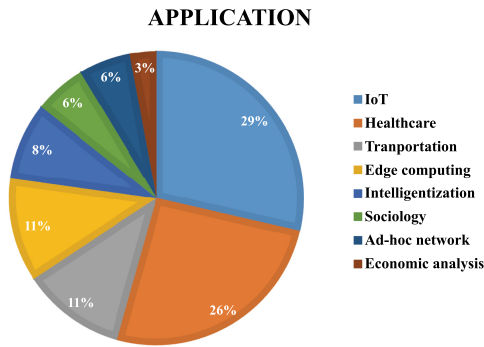


Fig. 9. Illustration of application scenarios applied in the literature (N=35).

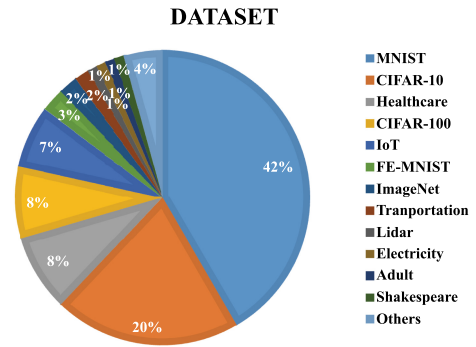


Fig. 10. Distributions of the dataset utilized in the literature (N=190).

encompasses smart cities, factories, and homes, represents 8%. Sociology and Ad-hoc network each has 6%, while Economic analysis accounts for 3% application.

Figure 9 also reveals that IoT becomes the leading application domain in DFL, driven by the rapid expansion of IoT devices. Healthcare and transportation rank second and third in the number of studies, with healthcare remaining important due to the sensitivity of medical data. In these two scenarios, numerous devices participate in training, where robustness and maintaining training accuracy despite attacks are vital considerations.

5.2 Datasets

To answer RQ3, Figure 10 illustrates the datasets used in the literature for experiments. This article investigates whether specific datasets are linked explicitly to particular application scenarios. The pie chart identifies the primary datasets and indicates whether the dataset pertains to image classification or natural language processing tasks.

As shown in Figure 10, MNIST [136] is the most widely used dataset at 42%, followed by CIFAR-10 [137] at 20%. MNIST includes the MNIST, Fashion-MNIST, and Extended-MNIST dataset. Healthcare-related datasets are used in 8%. FE-MNIST [138] is discussed in 3% of the cases. Depending on the research problem, various datasets are utilized for different scenarios, such as IoT, transportation, and lidar. The other category includes datasets uniquely mentioned in the filtered studies, such as the connect-4 [139], wikitext-2 [140], USPS [141], CINIC-10 [142], and KDD [143] datasets.

Figure 10 also reveals that over half the studies in the filtered studies utilize datasets for image classification tasks. LEAF [138] introduces the FE-MNIST dataset for FL to support Non-IID experiment settings. Healthcare datasets cover epilepsy [144], chest CT [145], PAMAP2 [146], COVID-19 [147], ISIC 2019 challenge dataset [148–150], Montgomery County X-Ray set [151], and Shenzhen Hospital X-ray set [152]. Most studies still use image classification for experiments, while rare studies investigate datasets related to natural language processing tasks. Datasets related to IoT or IoV are not the majority although some studies mention a connection between the proposed methods and IoT applications.

5.3 Training Models

In response to the RQ4, Figure 11 depicts the graph of neural network models evaluated in the literature. This article examines the neural networks employed in experiments with the previously mentioned datasets, particularly focusing on whether different studies use various models for image classification tasks.

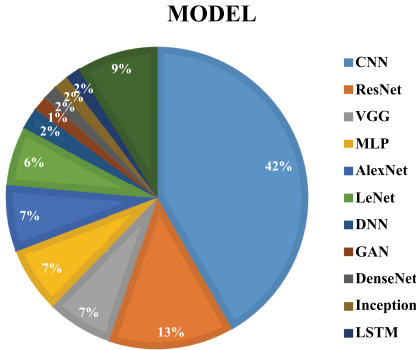


Fig. 11. Distributions of the neural network model employed in the literature ($N=127$).

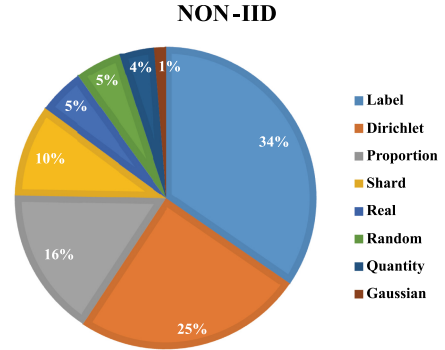


Fig. 12. Illustration of Non-IID settings for experiments ($N=81$).

As shown in Figure 11, **Convolutional Neural Network (CNN)** [153] is the most popular model at 42%, followed by **Residual Network (ResNet)** [154] at 13%. Model like VGG [155], **Multilayer Perception (MLP)** [156], AlexNet [157], and LeNet [153] share 7% for each. The Others category comprises neural networks mentioned only once across all filtered studies, such as **Spiking Neural Networks (SNNs)** [158, 159], **Gradient Boosting Decision Tree (GBDT)**, PointNet [160], regression models, mobilenet-v2 [161], mobilevit [162], transformer [163], logistic regression [164].

Figure 11 also reveals that most studies utilize image recognition datasets as a dominance of neural networks for image classification tasks. The pie chart reflects a significant portion of this trend, including CNN, ResNet, VGG, MLP, AlexNet, Inception [166], and **Densely Connected Convolutional Networks (DenseNet)** [167]. Moreover, CNN is a common model for image classification tasks due to its convenient design. **Long Short Term Memory (LSTM)** [165] is employed in studies focusing on text-related datasets, such as the Shakespeare dataset.

5.4 Non-IID Experiments

The goal of RQ5 in this article is to explore methods for generating diverse data distributions among clients under Non-IID settings. This article focuses on identifying the most commonly used Non-IID methods to determine if an optimal approach exists. Figure 12 depicts the strategies adopted in Non-IID experiments from the literature.

The label-based method is widely used in 34% of cases, where each node receives data with a fixed number of labels to generate diverse label distributions across nodes. The Dirichlet distribution [168], adopted by 25%, leads to inconsistent data distributions across clients. Proportion-based distribution, adopted by 16%, allocates data to nodes based on ratios to create varied distributions among nodes. The shard-based method, also adopted by 10%, splits the dataset into fixed shards for data distribution among training nodes. A few methods, such as real data and a random approach, account for 5% of each. Real data is used in medical institutions, where each institution trains on the collected data.

According to the filter rules set by this work, each qualifying study has to conduct a Non-IID experiment. However, not all studies elaborate on the implementation of a Non-IID environment. Moreover, some studies employ more than one Non-IID method for experiments. The label-based method is the most popular due to its ease of implementation. The practice of assigning a fixed set of labels to each node ensures that there is a variation in label distribution among the nodes.

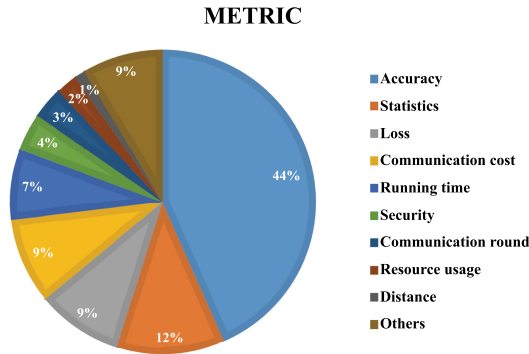


Fig. 13. Illustration of performance metrics conducted by the literature ($N=175$).

However, not all cases are suited for the label-based approach. For instance, in medical institutions, utilizing the local dataset collected by each institution for training is more appropriate than employing other segmentation methods. In addition, studies using the Dirichlet distribution for Non-IID experiments show an increasing trend in 2024.

5.5 Performance Metrics

To address RQ6 in this article, Figure 13 depicts the performance metrics conducted in the experiments. Moreover, this work aims to identify the most commonly used metrics and any unique performance metrics that arise due to specialized topics discussed in the literature. Note that a single study may evaluate multiple performance metrics. Therefore, Figure 13 counts on the total sum of metrics evaluated by the filtered studies.

As shown in Figure 13, accuracy is the most considered metric, accounting for 44% of all metrics. The second considered metric is the statistics, with a 12% usage rate. Statistics include various statistical measures such as **Root Mean Square Error (RMSE)**, **Mean Absolute Error (MAE)**, F1 Score, **Sum of Absolute Errors (SAE)**, Recall, Dice Score, and Geometric Mean. Loss and communication costs account for 9% of each. Communication cost involves traffic consumption and bandwidth usage per round. Running time is utilized in 7% of cases. Security-related and Resource-usage metrics account for 4% and 2%, respectively. Security-related metrics include **Deep Leakage from Gradients (DLG)**, backdoor success rate, accuracy drop, attack success rate, testing error rate, and image restore attack. The Resource usage tracks training resources such as CPU usage, Memory, Network Input Output, and Block Input Output. The distance metric [97] estimates the result between the proposed method and the baseline method. The Others category encloses various metrics mentioned only once in the filtered studies, such as Inception Score, Earth Mover's Distance, perplexity, contribution score, GAS, energy cost, pull request, and **Fréchet Inception Distance (FID)**.

After the SLR of performance metrics, this article notes that each study may have varied definitions of accuracy. Some studies calculate accuracy as an average over a local test dataset, while others use a global test dataset to estimate the accuracy. In discussions related to the Security-related category, many studies address accuracy declines due to attacks or the ability to reconstruct training data under parameter leaks. Communication rounds and running time have become more discussed in recent research, especially with a growing trend in running time. Accordingly, the time required for convergence in Non-IID experiments and additional costs incurred by enhancing DFL training security will receive more attention.

6 Findings and Future Challenges

This section discusses the key findings from the systematic review and presents future challenges in DFL. The discussion covers insights across architecture, topology, optimization, and security. Ongoing issues include Non-IID data distribution, limited diversity of datasets used in experiments, and evaluation inconsistencies. The analysis also proposes future research directions, such as the mitigation of communication overhead in large-scale model training, the development of an incentive mechanism to enable sustainable and practical deployment in real-world scenarios, and the advancement of DFL applications in privacy-sensitive domains like healthcare.

6.1 Clarifying DFL Definitions and Architectural Tradeoffs

During the SLR process, various definitions of DFL are proposed by different research works. According to the chronological order, the earliest study should be presented in the work [169], but later on, the study [170] is more recognized by other studies. Many studies claim to have proposed the DFL algorithm [87, 130] because of the ambiguity of the first DFL study. In our work, DFL refers to “Decentralized Federated Learning”, which means a FL system with a decentralized architecture. The DFL system eliminates a persistent centralized server to reduce the risk of a single point of failure. However, many studies incorrectly applied the terms “Decentralized Data” and “Distributed Data” to DFL. “Decentralized Data” and “Distributed Data” describe the characteristics of FL where data remains on the client side. Therefore, studies with these misunderstandings are excluded from the filter rules.

A comprehensive analysis across architecture, topology, optimization, and security perspectives reveals that the design of DFL systems is inherently shaped by task-specific tradeoffs and deployment constraints. Partially decentralized architectures balance aggregation role assignment and coordination efficiency. Fully decentralized frameworks focus on resilience and privacy but often increase communication complexity. Topological strategies serve different purposes. Clustering improves local coordination in heterogeneous settings. Structured designs such as ring and gossip protocols support efficient communication. Optimization techniques, including asynchronous training, model compression, and lightweight model exchanges, are used to enhance performance. These methods require careful calibration to avoid unstable convergence or reduced model quality. Mechanisms such as blockchain, differential privacy, and secure aggregation are integrated to enhance system integrity and data confidentiality. These techniques are particularly critical in high-risk applications such as healthcare and financial systems. Overall, these insights highlight the necessity for adaptive and modular DFL designs tailored to the operational characteristics of specific environments.

6.2 Challenges in Non-IID Experiments and Dataset Usages

In the study of Non-IID experiments, the challenge is to simulate the data distribution effectively. The label-based method is most used in the experiments. However, not all scenarios are applicable to this method. For example, datasets related to medical institutions are more suitably managed using the “real” approach. Each institution trains locally on the collected medical data, which is a more reasonable method than sharing the data across institutions. The “real” approach may also be applicable to IoV applications, where each vehicle collects and trains on a local dataset. Accordingly, our work suggests that each application should tailor approaches for Non-IID experiments. No single method is generally applicable to all application scenarios.

On the other hand, more than half of the studies utilized MNIST and CIFAR-10 datasets. Although many studies mentioned suitable applications for the proposed methods, most works still employed MNIST or CIFAR-10 for their experiments. The situation poses challenges in

discerning the effectiveness of the proposed methods on application-related datasets. Moreover, the proposed methods often utilize datasets of image classification tasks to measure the performance. Testing with different models, such as validating the adaptability of the proposed methods using **Recurrent Neural Network (RNN)** [171], is an option. Experiments with various models can demonstrate that the proposed method is versatile and not just effective for specific tasks. Furthermore, tasks related to natural language processing is also considered in the future. Only some of the collected studies discuss tasks associated with natural language processing in their experiments.

6.3 Performance Evaluations and Advanced Mechanisms

For the performance metrics, accuracy is one of the most widely used to measure empirical performance. However, the definition of accuracy varies across different studies. For instance, the source of the test dataset may be an 8:2 split by clients, or the system may share a global test dataset. No consistent standard exists for measuring accuracy due to varying definitions in each study. Apart from accuracy, many other performance metrics are further considered due to the main focus of each study. An increasing number of surveys are exploring metrics related to convergence time, such as convergence efficiency or the number of rounds needed to achieve a specific accuracy. Metrics related to resistance to attacks are also explored in security studies.

A recent study [187] indicates that traditional performance metrics may not fully capture the system-level performance of distributed systems. Conventional evaluations such as accuracy, latency, and convergence time mainly describe training results in centralized environments. DFL enables multiple clients to collaborate during training and may require broader evaluation metrics. The recent study suggests new metrics such as the adaptivity quotient and the fairness index. The adaptivity quotient measures how effectively a system reacts and recovers from node failures, network delays, or changes in data distribution. This metric reflects the stability and resilience of distributed training under dynamic conditions. The fairness index assesses the even distribution of computational resources among clients, preventing model bias or unfair collaboration caused by data imbalance, computational capacity, or communication frequency. These emerging metrics offer a different perspective on performance evaluation for decentralized federated learning.

Blockchain serves as a means to DFL due to its nature as a decentralized distributed ledger. Numerous studies have applied blockchain to FL to enhance security and privacy. Blockchain also addresses accuracy degradation caused by malicious nodes through novel mechanisms and regulates reward systems for training participants via smart contracts. Despite the advantages, integrating blockchain technology into DFL raises concerns, such as the potential need for additional computational resources for transaction processing and validation. Some studies discuss improving convergence efficiency in training and validation processes while applying blockchain. Beyond blockchain, studies exploring asynchronous mechanisms are rare. Asynchronous mechanisms mainly address straggler issues by significantly reducing the waiting time for communication. Exploring asynchronous mechanisms is a valuable direction for the future.

6.4 Future Research Opportunities in DFL

Future research may explore the integration of DFL with natural language processing tasks, especially in the context of a **Large Language Model (LLM)**. As LLMs typically require extensive training data and computational resources, applying DFL can help decentralize training across multiple institutions or edge devices while preserving data privacy. This is particularly valuable in domains where language data is sensitive or proprietary, such as in healthcare, legal, or enterprise settings. A recent study [188] also indicates that message brokers serving as middleware between

DFL systems and LLM applications can provide semantic routing, content filtering, and reliable delivery. Integrating brokers into decentralized environments poses system-level challenges in maintaining low latency and consistent message ordering under high-load, asynchronous communication. At the model level, further challenges involve addressing the high communication cost of large models, ensuring consistency and convergence under Non-IID language distributions, and adapting existing optimization techniques to DFL environments.

Designing robust incentive mechanisms is critical to preventing free riding. Ensuring fair contribution becomes especially important in large-scale DFL environments, where participant behavior may vary. Future research can explore dynamic reward allocation strategies based on factors such as historical performance, model quality, and data utility. Mechanisms like reputation systems can help monitor the consistency and reliability of client contributions. Token incentives managed through blockchain can support reliable and transparent reward schemes. These approaches are essential for sustaining long-term collaboration and participation in decentralized learning networks.

The healthcare domain presents a high-impact application scenario for DFL. Preserving patient privacy and complying with data protection regulations are essential requirements. For example, brain tumor detection from **Magnetic Resonance Imaging (MRI)** and **Computed Tomography (CT)** scans [185] can benefit from DFL training schemes that avoid direct sharing of sensitive imaging data. Integrating DFL into such diagnostic frameworks allows multiple medical institutions to collaboratively train robust deep learning models while maintaining strict data locality. DFL is especially valuable for improving diagnostic accuracy across diverse populations and imaging devices.

7 Conclusion

FL is an emerging ML approach that considers data privacy while training to analyze various information generated by participating clients. FL effectively addresses data privacy by avoiding the explicit sharing of raw data to achieve better security. The decentralized architecture is further introduced to enhance the robustness of the FL system. However, the Non-IID problem gradually emerges due to decentralization. The imbalanced datasets in each node lead to suboptimal performance of the FL system. To understand the state-of-the-art techniques, this article conducts an SLR on the topic of adding Non-IID to DFL. As of the end of 2024, a total of 93 studies from six academic databases have been compiled for analysis. After applying the selection criteria, irrelevant articles are excluded and four taxonomies are identified: architecture, topology, optimization, and security. This survey thoroughly analyzes the proposed methods regarding these topics to address various issues in DFL.

Based on the comprehensive category and analysis, this article outlines the popular applications, identifies the most frequently used datasets, and exposes the types of training models commonly involved in the literature. Moreover, this article discusses various approaches for Non-IID experiments and concludes that each application should tailor the design of a Non-IID experiment. Each method is only suitable for some application scenarios. Our work also reveals the most widely used metrics for performance evaluation in the current studies and the trends in future performance metrics. In the findings and challenges sections, this article addresses questions related to the definition of DFL, summarizes its applications and experiments, and discusses the pros and cons of integrating blockchain technology into DFL. Our work grasps current trends for future researchers to facilitate our findings and challenges in the DFL literature. Building on these insights, this article also highlights several future directions, underscoring the ongoing need to address Non-IID challenges and enhance the practical robustness of DFL systems.

References

- [1] Michael I. Jordan and Tom M. Mitchell. 2015. Machine learning: Trends, perspectives, and prospects. *Science* 349, 6245 (2015), 255–260.
- [2] Mu Li, David G. Andersen, Alexander J. Smola, and Kai Yu. 2014. Communication efficient distributed machine learning with the parameter server. In *Advances in Neural Information Processing Systems*, Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, and K. Q. Weinberger (Eds.). Vol. 27, Curran Associates, Inc. Retrieved from https://proceedings.neurips.cc/paper_files/paper/2014/file/935ad074f32d1e8f085a143449894cdc-Paper.pdf
- [3] Paul Voigt and Axel Von dem Bussche. 2017. The eu general data protection regulation (gdpr). *A Practical Guide*, 1st Ed., Cham: Springer International Publishing 10, 3152676, 10–5555.
- [4] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson and Blaise Aguera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial Intelligence and Statistics*. PMLR, 1273–1282.
- [5] Qinbin Li, Yiqun Diao, Quan Chen and Bingsheng He. 2022. Federated learning on non-iid data silos: An experimental study. In *2022 IEEE 38th International Conference on Data Engineering (ICDE)*. IEEE, 965–978.
- [6] Hangyu Zhu, Jinjin Xu, Shiqing Liu, and Yaochu Jin. 2021. Federated learning on non-IID data: A survey. *Neurocomputing* 465, 371–390.
- [7] Zili Lu, Heng Pan, Yueyue Dai, Xueming Si and Yan Zhang. 2024. Federated learning with non-iid data: A survey. *IEEE Internet of Things Journal* 11, 11 (2024), 19188–19209. DOI: [10.1109/JIOT.2024.3376548](https://doi.org/10.1109/JIOT.2024.3376548)
- [8] Enrique Tomás Martínez Beltrán, Mario Quiles Pérez, Pedro Miguel Sánchez Sánchez, Sergio López Bernal, Gérôme Bovet, Manuel Gil Pérez, Gregorio Martínez Pérez and Alberto Huertas Celdrán. 2023. Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges. *IEEE Communications Surveys & Tutorials* 25, 4 (2023), 2983–3013. DOI: [10.1109/COMST.2023.3315746](https://doi.org/10.1109/COMST.2023.3315746)
- [9] Laveen Bhatia and Saeed Samet. 2022. Decentralized federated learning: A comprehensive survey and a new blockchain-based data evaluation scheme. In *2022 Fourth International Conference on Blockchain Computing and Applications (BCCA)*. IEEE, 289–296.
- [10] Haoran Zhang, Shan Jiang, and Shichang Xuan. 2024. Decentralized federated learning based on blockchain: concepts, framework, and challenges. *Computer Communications* 216 (2024), 140–150. DOI: [10.1016/j.comcom.2023.12.042](https://doi.org/10.1016/j.comcom.2023.12.042)
- [11] Ehsan Hallaji, Roozbeh Razavi-Far, Mehrdad Saif, Boyu Wang, and Qiang Yang. 2024. Decentralized federated learning: A survey on security and privacy. *IEEE Transactions on Big Data* 10, 2 (2024), 194–213. DOI: [10.1109/TBDATA.2024.3362191](https://doi.org/10.1109/TBDATA.2024.3362191)
- [12] Liangqi Yuan, Ziran Wang, Lichao Sun, S Yu Philip, and Christopher G Brinton. 2024. Decentralized federated learning: A survey and perspective. *IEEE Internet of Things Journal* 11, 21 (2024), 34617–34638. DOI: [10.1109/JIOT.2024.3407584](https://doi.org/10.1109/JIOT.2024.3407584)
- [13] Staffs Keele. 2007. *Guidelines for performing systematic literature reviews in software engineering*. Technical report, ver. 2.3 ebse technical report. Ebse.
- [14] Barbara Kitchenham, O. Pearl Brereton, David Budgen, Mark Turner, John Bailey, and Stephen Linkman. 2009. Systematic literature reviews in software engineering—a systematic literature review. *Information and Software Technology* 51, 1 (2009), 7–15. DOI: [10.1016/j.infsof.2008.09.009](https://doi.org/10.1016/j.infsof.2008.09.009)
- [15] Li Li, Yuxi Fan, Mike Tse, and Kuo-Yi Lin. 2020. A review of applications in federated learning. *Computers & Industrial Engineering* 149 (2020), 106854. DOI: [10.1016/j.cie.2020.106854](https://doi.org/10.1016/j.cie.2020.106854)
- [16] Chen Zhang, Yu Xie, Hang Bai, Bin Yu, Weihong Li, and Yuan Gao. 2021. A survey on federated learning. *Knowledge-Based Systems* 216 (2021), 106775. DOI: [10.1016/j.knosys.2021.106775](https://doi.org/10.1016/j.knosys.2021.106775)
- [17] Mohammed Aledhari, Rehman Razzak, Reza M Parizi, and Fahad Saeed. 2020. Federated learning: A survey on enabling technologies, protocols, and applications. *IEEE Access* 8 (2020), 140699–140725. DOI: [10.1109/ACCESS.2020.3013541](https://doi.org/10.1109/ACCESS.2020.3013541)
- [18] Virraji Mothukuri, Reza M Parizi, Seyedamin Pouriyeh, Yan Huang, Ali Dehghantanha, and Gautam Srivastava. 2021. A survey on security and privacy of federated learning. *Future Generation Computer Systems* 115 (2021), 619–640. DOI: [10.1016/j.future.2020.10.007](https://doi.org/10.1016/j.future.2020.10.007)
- [19] Rémi Gosselin, Loïc Vieu, Faiza Loukil, and Alexandre Benoit. 2022. Privacy and security in federated learning: A survey. *Applied Sciences* 12, 19 (2022), 9901.
- [20] Attia Qammar, Ahmad Karim, Huansheng Ning, and Jianguo Ding. 2023. Securing federated learning with blockchain: a systematic literature review. *Artificial Intelligence Review* 56, 5 (2023), 3951–3985.
- [21] Xiaodong Ma, Jia Zhu, Zhihao Lin, Shanxuan Chen, and Yangjie Qin. 2022. A state-of-the-art survey on solving non-IID data in Federated Learning. *Future Generation Computer Systems* 135 (2022), 244–258. DOI: [10.1016/j.future.2022.05.003](https://doi.org/10.1016/j.future.2022.05.003)

- [22] Zihao Zhao, Yuzhu Mao, Yang Liu, Linqi Song, Ye Ouyang, Xinlei Chen, and Wenbo Ding. 2023. Towards efficient communications in federated learning: A contemporary survey. *Journal of the Franklin Institute*.
- [23] Omair Rashed Abdulwareth Almanifi, Chee-Onn Chow, Mau-Luen Tham, Joon Huang Chuah, and Jeevan Kanesan. 2023. Communication and computation efficiency in Federated Learning: A survey. *Internet of Things* 22 (2023), 100742. DOI: [10.1016/j.iot.2023.100742](https://doi.org/10.1016/j.iot.2023.100742)
- [24] Kilian Pfeiffer, Martin Rapp, Ramin Khalili, and Jörg Henkel. 2023. Federated learning for computationally-constrained heterogeneous devices: A survey. *ACM Computing Surveys* 55, 14s, Article 334 (July 2023), 27. DOI: [10.1145/3596907](https://doi.org/10.1145/3596907)
- [25] Xiao Wang, Lina Ge and Guifeng Zhang. 2023. A review of client selection mechanisms in heterogeneous federated learning. In *International Conference on Intelligent Computing*. Springer, 761-772.
- [26] Chenhao Xu, Youyang Qu, Yong Xiang, and Longxiang Gao. 2023. Asynchronous federated learning on heterogeneous devices: A survey. *Computer Science Review* 50 (2023), 100595. DOI: [10.1016/j.cosrev.2023.100595](https://doi.org/10.1016/j.cosrev.2023.100595)
- [27] Behnaz Soltani, Venus Haghighi, Adnan Mahmood, Quan Z Sheng, and Lina Yao. 2022. A survey on participant selection for federated learning in mobile networks. In *Proceedings of the 17th ACM Workshop on Mobility in the Evolving Internet Architecture*. 19–24.
- [28] Lei Fu, Huanle Zhang, Ge Gao, Mi Zhang, and Xin Liu. 2023. Client selection in federated learning: Principles, challenges, and opportunities. *IEEE Internet of Things Journal* 10, 24 (2023), 21811–21819. DOI: [10.1109/JIOT.2023.3299573](https://doi.org/10.1109/JIOT.2023.3299573)
- [29] Zhongchang Zhou, Fenggang Sun, Xiangyu Chen, Dongxu Zhang, Tianzhen Han, and Peng Lan. 2023. A decentralized federated learning based on node selection and knowledge distillation. *Mathematics* 11, 14 (2023), 3162. DOI: [10.3390/math11143162](https://doi.org/10.3390/math11143162)
- [30] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. arXiv:1503.02531. Retrieved from <https://arxiv.org/abs/1503.02531>
- [31] Helin Yang, Kwok-Yan Lam, Liang Xiao, Zehui Xiong, Hao Hu, Dusit Niyato, and H Vincent Poor. 2022. Lead federated neuromorphic learning for wireless edge artificial intelligence. *Nature Communications* 13, 1 (2022), 4269.
- [32] Renuga Kanagavelu, Qingsong Wei, Zengxiang Li, Haibin Zhang, Juniarto Samsudin, Yechao Yang, Rick Siow Mong Goh, and Shanguang Wang. 2022. CE-Fed: Communication efficient multi-party computation enabled federated learning. *Array* 15 (2022), 100207. DOI: [10.1016/j.array.2022.100207](https://doi.org/10.1016/j.array.2022.100207)
- [33] Qianpiao Ma, Jianchun Liu, Hongli Xu, Qingmin Jia, and Renchao Xie. 2024. FRACTAL: Data-aware clustering and communication optimization for decentralized federated learning. *IEEE Transactions on Big Data*.
- [34] Zhen Qin, Xueqiang Yan, Mengchu Zhou, and Shuiguang Deng. 2024. BlockDFL: A Blockchain-based Fully Decentralized Peer-to-Peer Federated Learning Framework. In *Proceedings of the ACM on Web Conference 2024*. 2914–2925.
- [35] Yuta Tomimasu and Koya Sato. 2023. On adaptive client/miner selection for efficient blockchain-based decentralized federated learning. In *2023 IEEE 98th Vehicular Technology Conference (VTC2023-Fall)*. IEEE, 1–6.
- [36] Miao Yang, Ximin Wang, Hongbin Zhu, Haifeng Wang, and Hua Qian. 2021. Federated learning with class imbalance reduction. In *2021 29th European Signal Processing Conference (EUSIPCO)*. IEEE, 2174–2178.
- [37] Bo Cui and Tianyu Mei. 2023. ABFL: A blockchain-enabled robust framework for secure and trustworthy federated learning. In *Proceedings of the 39th Annual Computer Security Applications Conference*. 636–646.
- [38] Gimoon Woo, Hyungbin Kim, Seunghyun Park, Cheolwoo You, and Hyunhee Park. 2022. Fairness-based multi-AP Coordination Using Federated Learning in Wi-Fi 7. *Sensors* 22, 24 (2022), 9776.
- [39] Zhao Wang, Yifan Hu, Shiyang Yan, Zhihao Wang, Ruijie Hou and Chao Wu. 2022. Efficient ring-topology decentralized federated learning with deep generative models for medical data in ehealthcare systems. *Electronics* 11, 10 (2022), 1548.
- [40] Pitch Patarasuk and Xin Yuan. 2009. Bandwidth optimal all-reduce algorithms for clusters of workstations. *Journal of Parallel and Distributed Computing* 69, 2 (2009), 117–124.
- [41] Rajeev Thakur, Rolf Rabenseifner, and William Gropp. 2005. Optimization of collective communication operations in MPICH. *The International Journal of High Performance Computing Applications* 19, 1 (2005), 49–66.
- [42] Zheyi Chen, Weixian Liao, Pu Tian, Qianlong Wang, and Wei Yu. 2022. A fairness-aware peer-to-peer decentralized learning framework with heterogeneous devices. *Future Internet* 14, 5 (2022), 138.
- [43] Shivam Kalra, Junfeng Wen, Jesse C. Cresswell, Maksims Volkovs, and HR Tizhoosh. 2023. Decentralized federated learning through proxy model sharing. *Nature communications* 14, 1 (2023), 2899.
- [44] Changji Wang, Canjie Pan, Qingqing Gan, Zhen Liu, Ning Liu, and Jingjing Zhang. 2024. FedSCD: Federated learning with semi-centralization, discrepancy-awareness and dual-model collaboration. In *Australasian Conference on Information Security and Privacy*. Springer, 174–192.
- [45] Mahmoud Assran, Nicolas Loizou, Nicolas Ballas, and Mike Rabbat. 2019. Stochastic gradient push for distributed deep learning. In *International Conference on Machine Learning*. PMLR, 344–353.

- [46] Qingzhe Jin and Hideya Ochiai. 2022. Decentralized P2P federated learning on ad-hoc like networks with Non-IID dataset. In *2022 18th International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob)*. IEEE, 387–393.
- [47] Matteo Pennisi, Federica Proietto Salanitri, Giovanni Bellitto, Concetto Spampinato, Simone Palazzo, Bruno Casella and Marco Aldinucci. 2023. Experience replay as an effective strategy for optimizing decentralized federated learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 3376–3383.
- [48] Danya Xu, Mingwei Jia, Tao Chen, Yi Liu, Tianyou Chai, and Tao Yang. 2024. Decentralized federated domain generalization with cluster alignment for fault diagnosis. *Control Engineering Practice* 148 (2024), 105951. DOI: [10.1016/j.conengprac.2024.105951](https://doi.org/10.1016/j.conengprac.2024.105951)
- [49] Wenxiao Lou, Yang Xu, Hongli Xu, and Yunming Liao. 2023. Decentralized federated learning with data feature transmission and neighbor selection. In *2022 IEEE 28th International Conference on Parallel and Distributed Systems (ICPADS)*. IEEE, 688–695.
- [50] Renuga Kanagavelu, Chris George Anil, Yuan Wang, Huazhu Fu, Qingsong Wei, Yong Liu, and Rick Siow Mong Goh. 2024. Fed-SHARC: Resilient decentralized federated learning based on reward driven clustering. In *2024 IEEE Conference on Artificial Intelligence (CAI)*. IEEE, 581–586.
- [51] Zhikun Chen, Daofeng Li, Jinkang Zhu, and Sihai Zhang. 2022. DACFL: Dynamic average consensus-based federated learning in decentralized sensors network. *Sensors* 22, 9 (2022), 3317.
- [52] Minghui Zhu and Sonia Martinez. 2010. Discrete-time dynamic average consensus. *Automatica* 46, 2 (2010), 322–329.
- [53] Lun Wang, Yang Xu, Hongli Xu, Min Chen, and Liusheng Huang. 2023. Accelerating decentralized federated learning in heterogeneous edge computing. *IEEE Transactions on Mobile Computing* 22, 9 (2023), 5001–5016. DOI: [10.1109/TMC.2022.3178378](https://doi.org/10.1109/TMC.2022.3178378)
- [54] Xiao Zhang, Yangyang Wang, Shuzhen Chen, Cui Wang, Dongxiao Yu, and Xiuzhen Cheng. 2023. Robust communication-efficient decentralized learning with heterogeneity. *Journal of Systems Architecture* 141 (2023), 102900. DOI: [10.1016/j.sysarc.2023.102900](https://doi.org/10.1016/j.sysarc.2023.102900)
- [55] Min Chen, Yang Xu, Hongli Xu, and Liusheng Huang. 2023. Enhancing decentralized federated learning for non-iid data on heterogeneous devices. In *2023 IEEE 39th International Conference on Data Engineering (ICDE)*. IEEE, 2289–2302.
- [56] Yunchuan Shi, Wei Li, Xiaomin Chang, Ting Yang, Yaojie Sun, and Albert Y Zomaya. 2023. On enabling collaborative non-intrusive load monitoring for sustainable smart cities. *Scientific Reports* 13, 1 (2023), 6569.
- [57] Ion Stoica, Robert Morris, David Liben-Nowell, David R Karger, M Frans Kaashoek, Frank Dabek, and Hari Balakrishnan. 2003. Chord: a scalable peer-to-peer lookup protocol for internet applications. *IEEE/ACM Transactions on Networking* 11, 1 (2003), 17–32.
- [58] Jianjun Huang, Zihao Rui, and Li Kang. 2024. Fedisp: an incremental subgradient-proximal-based ring-type architecture for decentralized federated learning. *Complex & Intelligent Systems* 10, 2 (1 Apr 2024), 2499–2514. DOI: [10.1007/s40747-023-01272-4](https://doi.org/10.1007/s40747-023-01272-4)
- [59] Rongxu Xu, Wenquan Jin, Anam Nawaz Khan, Sunhwan Lim, and Do-Hyeun Kim. 2023. Cooperative swarm learning for distributed cyclic edge intelligent computing. *Internet of Things* 22 (2023), 100783. DOI: [10.1016/j.iot.2023.100783](https://doi.org/10.1016/j.iot.2023.100783)
- [60] Zhenheng Tang, Shaohuai Shi, Bo Li and Xiaowen Chu. 2022. GossipFL: A decentralized federated learning framework with sparsified and adaptive communication. *IEEE Transactions on Parallel and Distributed Systems* 34, 3 (2022), 909–922.
- [61] Alan Demers, Dan Greene, Carl Hauser, Wes Irish, John Larson, Scott Shenker, Howard Sturgis, Dan Swinehart, and Doug Terry. 1987. Epidemic algorithms for replicated database maintenance. In *Proceedings of the Sixth Annual ACM Symposium on Principles of Distributed Computing*. 1–12.
- [62] Xiaojun Cai, Nanxiang Yu, Mengying Zhao, Mei Cao, Tingting Zhang, and Jianbo Lu. 2024. Decentralized federated learning in partially connected networks with Non-IID data. In *2024 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 1–6.
- [63] Suo Chen, Yang Xu, Hongli Xu, Zhenguo Ma, and Zhiyuan Wang. 2024. Enhancing decentralized and personalized federated learning with topology construction. *IEEE Transactions on Mobile Computing* 23, 10 (2024), 9692–9707. DOI: [10.1109/TMC.2024.3367872](https://doi.org/10.1109/TMC.2024.3367872)
- [64] Siyuan Liu, Zhiqiang Liu, Zhiwei Xu, Wenjing Liu, and Jie Tian. 2023. Hierarchical decentralized federated learning framework with adaptive clustering: Bloom-filter-based companions choice for learning non-IID data in IoT. *Electronics* 12, 18 (2023), 3811.
- [65] Wenjing Liu, Zhiwei Xu, Jie Tian and Yujun Zhang. 2021. Towards In-Network compact representation: Mergeable counting bloom filter vis cuckoo scheduling. *IEEE Access* 9 (2021), 55329–55339.
- [66] Sally M El-Ghamrawy. 2016. A knowledge management framework for imbalanced data using frequent pattern mining based on bloom filter. In *2016 11th International Conference on Computer Engineering & Systems (ICCES)*. IEEE, 226–231.

- [67] Yanbin Li, Xuemei Wang, Runkang Sun, Xiaojun Xie, Shijia Ying, and Shougang Ren. 2023. Trustiness-based hierarchical decentralized federated learning. *Knowledge-Based Systems* 276 (2023), 110763. DOI : [10.1016/j.knosys.2023.110763](https://doi.org/10.1016/j.knosys.2023.110763)
- [68] Yang Lu, Zhengxin Yu, and Neeraj Suri. 2023. Privacy-preserving decentralized federated learning over time-varying communication graph. *ACM Transactions on Privacy and Security* 26, 3 (2023), 1–39.
- [69] Lin Xiao, Stephen Boyd, and Seung-Jean Kim. 2007. Distributed average consensus with least-mean-square deviation. *Journal of Parallel and Distributed Computing* 67, 1 (2007), 330–46.
- [70] Zhipeng Gao, Huangqi Li, Yijing Lin, Ze Chai, Yang Yang, and Lanlan Rui. 2022. Effective blockchain-based asynchronous federated learning for edge-computing. In *International Conference on Collaborative Computing: Networking, Applications and Worksharing*. Springer, 514–532.
- [71] Yuhao Zhou, Minjia Shi, Yuxin Tian, Qing Ye, and Jiancheng Lv. 2024. DeFTA: A plug-and-play peer-to-peer decentralized federated learning framework. *Information Sciences* 670 (2024), 120582. DOI : [10.1016/j.ins.2024.120582](https://doi.org/10.1016/j.ins.2024.120582)
- [72] Yongding Tian, Zhuoran Guo, Jiaxuan Zhang, and Zaid Al-Ars. 2023. Dfl: High-performance blockchain-based federated learning. *Distributed Ledger Technologies: Research and Practice* 2, 3 (2023), 1–25.
- [73] Hong Zhu, Lisha Gao, Yitian Sha, Nan Xiang, Yue Wu, and Shuo Han. 2023. Decentralized heterogeneous federal distillation learning based on blockchain. *Computers, Materials & Continua* 76, 3.
- [74] Jun Liu, Jianchun Liu, Hongli Xu, Yunming Liao, Zhiyuan Wang, and Qianpiao Ma. 2024. YOGA: Adaptive layer-wise model aggregation for decentralized federated learning. *IEEE/ACM Transactions on Networking* 32, 2 (2024), 1768–1780. DOI : [10.1109/TNET.2023.3329005](https://doi.org/10.1109/TNET.2023.3329005)
- [75] Chengxi Li, Gang Li, and Pramod K Varshney. 2021. Decentralized federated learning via mutual knowledge transfer. *IEEE Internet of Things Journal* 9, 2 (2021), 1136–1147.
- [76] Ying Zhang, Tao Xiang, Timothy M Hospedales, and Huchuan Lu. 2018. Deep mutual learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 4320–4328.
- [77] Yikuan Chen, Li Liang, and Wei Gao. 2023. DFedSN: Decentralized federated learning based on heterogeneous data in social networks. *World Wide Web* 26, 5 (1 Sep 2023), 2545–2568. DOI : [10.1007/s11280-023-01152-4](https://doi.org/10.1007/s11280-023-01152-4)
- [78] Amirhossein Reisizadeh, Farzan Farnia, Ramtin Pedarsani, and Ali Jadbabaie. 2020. Robust federated learning: The case of affine distribution shifts. *Advances in Neural Information Processing Systems* 33, 21554–21565.
- [79] Jossekin Beilharz, Bjarne Pfitzner, Robert Schmid, Paul Geppert, Bert Arnrich, and Andreas Polze. 2021. Implicit model specialization through dag-based decentralized federated learning. In *Proceedings of the 22nd International Middleware Conference*. 310–322.
- [80] Xin Zhang, Minghong Fang, Zhuqing Liu, Haibo Yang, Jia Liu, and Zhengyuan Zhu. 2022. Net-fleet: Achieving linear convergence speedup for fully decentralized federated learning with heterogeneous data. In *Proceedings of the Twenty-Third International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing*. 71–80.
- [81] Ye Li, Jiale Zhang, Junwu Zhu, and Wenjuan Li. 2022. HBMD-FL: Heterogeneous federated learning algorithm based on blockchain and model distillation. In *International Symposium on Emerging Information Security and Applications*. Springer, 145–159.
- [82] Saleh Baghersalimi, Tomas Teijeiro, Amir Aminifar, and David Atienza. 2024. Decentralized federated learning for epileptic seizures detection in low-power wearable systems. *IEEE Transactions on Mobile Computing* 23, 5 (2024), 6392–6407. DOI : [10.1109/TMC.2023.3320862](https://doi.org/10.1109/TMC.2023.3320862)
- [83] TV Nguyen, MA Dakka, SM Diakiw, MD VerMilyea, M Perugini, JMM Hall, and D Perugini. 2022. A novel decentralized federated learning approach to train on globally distributed, poor quality, and protected private medical data. *Scientific Reports* 12, 1 (2022), 8888.
- [84] Huanhuan Ge. 2024. Extending knowledge distillation for personalized federation. In *International Conference on Intelligent Computing*. Springer, 392–403.
- [85] Kota Maejima, Takayuki Nishio, Asato Yamazaki, and Yuko Hara-Azumi. 2024. Tram-FL: Routing-based model training for decentralized federated learning. In *2024 IEEE 21st Consumer Communications & Networking Conference (CCNC)*. IEEE, 1038–1039.
- [86] Eunjeong Jeong and Marios Kountouris. 2023. Personalized decentralized federated learning with knowledge distillation. In *ICC 2023 - IEEE International Conference on Communications*. IEEE, 1982–1987.
- [87] Alessandro Giuseppe, Sabato Manfredi, and Antonio Pietrabissa. 2022. A weighted average consensus approach for decentralized federated learning. *Machine Intelligence Research* 19, 4 (2022), 319–330.
- [88] Michael Duchesne, Kaiwen Zhang, and Talhi Chamseddine. 2024. Multi-Confederated learning: Inclusive Non-IID data handling with decentralized federated learning. In *Proceedings of the 39th ACM/SIGAPP Symposium on Applied Computing*. 1587–1595.
- [89] Agnideven Palanisamy Sundar, Feng Li, Xukai Zou, and Tianchong Gao. 2024. Subjective logic-based decentralized federated learning for Non-IID data. In *Proceedings of the 19th International Conference on Availability, Reliability and Security*. 1–11.

- [90] Audun Jøsang. 1997. Artificial reasoning with subjective logic. In *Proceedings of the Second Australian Workshop on Commonsense Reasoning*. Citeseer, 34.
- [91] Shivam Kalra, Junfeng Wen, Jesse C Cresswell, Maksims Volkovs, and Hamid R Tizhoosh. 2023. Decentralized federated learning through proxy model sharing. *Nature communications* 14, 1 (2023), 2899.
- [92] Lars Wulfert, Navidreza Asadi, Wen-Yu Chung, Christian Wiede, and Anton Grabmaier. 2023. Adaptive decentralized federated gossip learning for resource-constrained IoT devices. In *Proceedings of the 4th International Workshop on Distributed Machine Learning*. 27–33.
- [93] Su Wang, Seyyedali Hosseinalipour, and Christopher G Brinton. 2024. Multi-source to multi-target decentralized federated domain adaptation. *IEEE Transactions on Cognitive Communications and Networking*.
- [94] Da Wu, Xiuhuan Zang, Jiewei Chen, Xinpeng Wu, Yu Lu, and Feng Qi. 2023. An efficient decentralized federated learning framework based on directed acyclic graph. In *International Conference on Computer Engineering and Networks*. Springer, 209–220.
- [95] Vaikkunth Mugunthan, Ravi Rahman, and Lalana Kagal. 2022. BlockFlow: Decentralized, privacy-preserving, and accountable federated machine learning. In *Blockchain and Applications: 3rd International Congress*. Springer, 233–242.
- [96] Andras Ferenczi and Costin Bădică. 2023. A fully decentralized privacy-enabled federated learning system. In *International Conference on Computational Collective Intelligence*. Springer, 444–456.
- [97] Ziwen Cheng, Yi Liu, Chao Wu, Yongqi Pan, Liushun Zhao, and Cheng Zhu. 2023. PoShapley-BCFL: A fair and robust decentralized federated learning based on blockchain and the proof of shapley-value. In *International Conference on Neural Information Processing*. Springer, 531–549.
- [98] Eisuke Tomiyama, Hiroshi Esaki, and Hideya Ochiai. 2023. Competitive and asynchronous decentralized federated learning with blockchain smart contracts. In *Proceedings of the 2023 ACM Conference on Information Technology for Social Good*. 92–99.
- [99] Tao Chen, Xiaofen Wang, and Hong-Ning Dai. 2024. R-PPDFL: A robust and privacy-preserving decentralized federated learning system. In *Australasian Conference on Information Security and Privacy*. Springer, 158–173.
- [100] Juan Benet. 2014. IpfS-content addressed, versioned, p2p file system. arXiv:1407.3561. Retrieved from <https://arxiv.org/abs/1407.3561>
- [101] Gavin Wood. 2014. Ethereum: A secure decentralised generalised transaction ledger. *Ethereum Project Yellow Paper* 151, (2014), 1–32.
- [102] Jun Li, Yumeng Shao, Kang Wei, Ming Ding, Chuan Ma, Long Shi, Zhu Han, and H Vincent Poor. 2021. Blockchain assisted decentralized federated learning (BLADE-FL): Performance analysis and resource allocation. *IEEE Transactions on Parallel and Distributed Systems* 33, 10 (2021), 2401–2415.
- [103] Asato Yamazaki, Takayuki Nishio, and Yuko Hara-Azumi. 2023. Convergence improvement by parameters exchange in asynchronous decentralized federated learning for Non-IID data. In *2023 49th Euromicro Conference on Software Engineering and Advanced Applications (SEAA)*. IEEE, 36–43.
- [104] Mingyang Zhou, Gang Liu, KeZhong Lu, Rui Mao, and Hao Liao. 2024. Accelerating the decentralized federated learning via manipulating edges. In *Proceedings of the ACM on Web Conference 2024*. 2945–2954.
- [105] Dimitri P Bertsekas. 2011. Incremental gradient, subgradient, and proximal methods for convex optimization: A survey. *Optimization for Machine Learning* 1–38, 3.
- [106] Rohit Parasnis, Seyyedali Hosseinalipour, Yun-Wei Chu, Mung Chiang, and Christopher G Brinton. 2023. Connectivity-aware semi-decentralized federated learning over time-varying d2d networks. In *Proceedings of the Twenty-fourth International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing*. 31–40.
- [107] Yatong Wang, Yuncheng Wu, Xincheng Chen, Gang Feng, and Beng Chin Ooi. 2023. Incentive-Aware Decentralized Data Collaboration. *Proceedings of the ACM on Management of Data* 1, 2 (2023), 1–27.
- [108] Kaiqing Zhang, Zhuoran Yang, and Tamer Başar. 2021. Multi-agent reinforcement learning: A selective overview of theories and algorithms. *Handbook of Reinforcement Learning and Control*, 321–384.
- [109] Satoshi Nakamoto. 2008. Bitcoin: A peer-to-peer electronic cash system. *Decentralized Business Review*. Retrieved from <https://bitcoin.org/bitcoin.pdf>
- [110] CU Om Kumar, Sudhakaran Gajendran, V Balaji, A Nhveen, and S Sai Balakrishnan. 2023. Securing health care data through blockchain enabled collaborative machine learning. *Soft Computing* 27, 14 (2023), 9941–9954.
- [111] Djallel Hamouda, Mohamed Amine Ferrag, Nadjette Benhamida, and Hamid Seridi. 2023. PPSS: A privacy-preserving secure framework using blockchain-enabled federated deep learning for industrial IoTs. *Pervasive and Mobile Computing* 88 (2023), 101738. DOI: [10.1016/j.pmcj.2022.101738](https://doi.org/10.1016/j.pmcj.2022.101738)
- [112] Yipeng Dong, Lei Zhang, and Lin Xu. 2023. Privacy-preserving and reliable distributed federated learning. In *International Conference on Algorithms and Architectures for Parallel Processing*. Springer, 130–149.

- [113] Bing Wu and Haiyan Kang. 2023. Research on federated sharing methods for massive data in blockchain. In *International Conference on Smart Grid and Internet of Things*. Springer, 12–27.
- [114] Xiao Li, Jianhuan Zeng, Chen Chen, Teng Li, and Jun Ma. 2024. Decentralized adaptive work package learning for personalized and privacy-preserving occupational health and safety monitoring in construction. *Automation in Construction* 165, 105556.
- [115] Martin Abadi, Andy Chu, Ian Goodfellow, H Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. 2016. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security (Vienna, Austria) (CCS'16)*. Association for Computing Machinery, New York, NY, USA, 308–318. DOI: [10.1145/2976749.2978318](https://doi.org/10.1145/2976749.2978318)
- [116] Othmane Friha, Mohamed Amine Ferrag, Mohamed Benbouzid, Tarek Berghout, Burak Kantarci, and Kim-Kwang Raymond Choo. 2023. 2DF-IDS: Decentralized and differentially private federated learning-based intrusion detection system for industrial IoT. *Computers & Security* 127 (2023), 103097. DOI: [10.1016/j.cose.2023.103097](https://doi.org/10.1016/j.cose.2023.103097)
- [117] Qiuxian Li, Quanxing Zhou, Mingyang Li, and Zhenlong Wang. 2023. Trustworthy anti-collusion federated learning scheme optimized by game theory. *Electronics* 12, 18 (2023), 3867.
- [118] Mehrdad Nojoumian. 2018. Rational trust modeling. In *Proceedings Decision and Game Theory for Security: 9th International Conference, GameSec 2018*, Seattle, WA, USA, October 29–31, 2018, 9. Springer, 418–431.
- [119] Xuyang Ma, Du Xu, and Katinka Wolter. 2024. Towards blockchain-enabled decentralized and secure federated learning. *Information Sciences* 665, 120368.
- [120] Feng Yu, Hui Lin, Xiaoding Wang, Abdussalam Yassine, and M Shamim Hossain. 2022. Blockchain-empowered secure federated learning system: Architecture and applications. *Computer Communications* 196 (2022), 55–65. DOI: [10.1016/j.comcom.2022.09.008](https://doi.org/10.1016/j.comcom.2022.09.008)
- [121] Yuzheng Li, Chuan Chen, Nan Liu, Huawei Huang, Zibin Zheng and Qiang Yan. 2020. A blockchain-based decentralized federated learning framework with committee consensus. *IEEE Network* 35, 1 (2020), 234–241.
- [122] Sicong Zhou, Huawei Huang, Ruixin Li, Jialiang Liu, and Zibin Zheng. 2023. ComAvg: Robust decentralized federated learning with random committees. *Computer Communications* 211 (2023), 147–156. DOI: [10.1016/j.comcom.2023.09.004](https://doi.org/10.1016/j.comcom.2023.09.004)
- [123] Rachid Guerraoui and Sébastien Rouault. 2018. The hidden vulnerability of distributed learning in byzantium. In *International Conference on Machine Learning*. PMLR, 3521–3530.
- [124] Dong Yin, Yudong Chen, Ramchandran Kannan, and Peter Bartlett. 2018. Byzantine-robust distributed learning: Towards optimal statistical rates. In *International Conference on Machine Learning*. PMLR, 5650–5659.
- [125] Guangxi Lu, Zuobin Xiong, Ruinian Li, Nael Mohammad, Yingshu Li, and Wei Li. 2023. DEFEAT: A decentralized Federated Learning against gradient attacks. *High-Confidence Computing* 3, 3 (2023), 100128. DOI: [10.1016/j.hcc.2023.100128](https://doi.org/10.1016/j.hcc.2023.100128)
- [126] Matt Fredrikson, Somesh Jha, and Thomas Ristenpart. 2015. Model inversion attacks that exploit confidence information and basic countermeasures. In *Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security*. 1322–1333.
- [127] Stacey Truex, Ling Liu, Mehmet Emre Gursoy, Lei Yu and Wenqi Wei. 2019. Demystifying membership inference attacks in machine learning as a service. *IEEE Transactions on Services Computing* 14, 6 (2019), 2073–2089.
- [128] Zhongyuan Zhao, Lunchao Xia, Luyao Jiang, Wang Gao, and Quanbo Ge. 2024. Zeroth-order gradient tracking for decentralized learning with privacy guarantees. *ISA Transactions* 152 (2024), 1–14. DOI: [10.1016/j.isatra.2024.06.033](https://doi.org/10.1016/j.isatra.2024.06.033)
- [129] Zhongyuan Zhao, Ju Yang, Wang Gao, Yan Wang, and Mengli Wei. 2024. Differentially private distributed online optimization via push-sum one-point bandit dual averaging. *Neurocomputing* 572 (2024), 127184. DOI: [10.1016/j.neucom.2023.127184](https://doi.org/10.1016/j.neucom.2023.127184)
- [130] Guangxi Lu, Zuobin Xiong, Ruinian Li, and Wei Li. 2022. Decentralized federated learning: A defense against gradient inversion attack. In *International Wireless Internet Conference*. Springer, 44–56.
- [131] Torben Pryds Pedersen. 1991. Non-interactive and information-theoretic secure verifiable secret sharing. In *Annual International Cryptology Conference*. Springer, 129–140.
- [132] Yingchun Cui and Jinghua Zhu. 2023. Privacy preserving federated learning framework based on multi-chain aggregation. In *International Conference on Database Systems for Advanced Applications*. Springer, 693–702.
- [133] Adi Shamir. 1979. How to share a secret. *Communications of the ACM* 22, 11 (1979), 612–613.
- [134] Yikuan Chen, Li Liang, and Wei Gao. 2023. Non trust detection of decentralized federated learning based on historical gradient. *Engineering Applications of Artificial Intelligence* 120 (2023), 105888. DOI: [10.1016/j.engappai.2023.105888](https://doi.org/10.1016/j.engappai.2023.105888)
- [135] Ashish Gupta, Tie Luo, Mao V Ngo, and Sajal K Das. 2022. Long-short history of gradients is all you need: Detecting malicious and unreliable clients in federated learning. In *European Symposium on Research in Computer Security*. Springer, 445–465.
- [136] L. Deng. 2012. The MNIST database of handwritten digit images for machine learning research [Best of the Web]. *IEEE Signal Processing Magazine* 29, 6 (2012), 141–142.

- [137] Alex Krizhevsky, Geoffrey Hinton, et al. 2009. Learning multiple layers of features from tiny images.
- [138] Sebastian Caldas, Sai Meher Karthik Duddu, Peter Wu, Tian Li, Jakub Konečný, H Brendan McMahan, Virginia Smith, and Ameet Talwalkar. 2018. Leaf: A benchmark for federated settings. arXiv:1812.01097. Retrieved from <https://arxiv.org/abs/1812.01097>
- [139] John Tromp. 1995. Connect-4. *UCI Machine Learning Repository*. DOI : <https://doi.org/10.24432/C59P43>
- [140] Stephen Merity, Caiming Xiong, James Bradbury and Richard Socher. 2016. Pointer sentinel mixture models. arXiv:1609.07843. Retrieved from <https://arxiv.org/abs/1609.07843>
- [141] Jonathan J. Hull. 1994. A database for handwritten text recognition research. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 16, 5 (1994), 550–554.
- [142] Luke N Darlow, Elliot J Crowley, Antreas Antoniou, and Amos J Storkey. 2018. Cinic-10 is not imagenet or cifar-10. arXiv:1810.03505. Retrieved from <https://arxiv.org/abs/1810.03505>
- [143] Fan Wei Lee Wenke Prodromidis Andreas Stolfo, Salvatore and Philip Chan. 1999. KDD Cup 1999 Data. UCI Machine Learning Repository. DOI : <https://doi.org/10.24432/C51C7N>
- [144] Vinit Shah, Eva Von Weltin, Silvia Lopez, James Riley McHugh, Lillian Veloso, Meysam Golmohammadi, Iyad Obeid, and Joseph Picone. 2018. The temple university hospital seizure detection corpus. *Frontiers in Neuroinformatics* 12 - 2018 (2018). DOI : [10.3389/fninf.2018.00083](https://doi.org/10.3389/fninf.2018.00083)
- [145] Rachel Lea Draelos, David Dov, Maciej A. Mazurowski, Joseph Y. Lo, Ricardo Henao, Geoffrey D. Rubin, and Lawrence Carin. 2021. Machine-learning-based multiple abnormality prediction with large-scale chest computed tomography volumes. *Medical Image Analysis* 67 (2021), 101857.
- [146] Attila Reiss and Didier Stricker. 2012. Introducing a new benchmarked dataset for activity monitoring. In *2012 16th International Symposium on Wearable Computers*. IEEE, 108–109.
- [147] Xingyi Yang, Xuehai He, Jinyu Zhao, Yichen Zhang, Shanghang Zhang and Pengtao Xie. 2020. COVID-CT-dataset: A CT scan dataset about COVID-19. arXiv:2003.13865. Retrieved from <https://arxiv.org/abs/2003.13865>
- [148] Noel CF Codella, David Gutman, M Emre Celebi, Brian Helba, Michael A Marchetti, Stephen W Dusza, Aadi Kallou, Konstantinos Liopyris, Nabin Mishra, and Harald Kittler. 2018. Skin lesion analysis toward melanoma detection: A challenge at the 2017 international symposium on biomedical imaging (isbi), hosted by the international skin imaging collaboration (isic). In *2018 IEEE 15th International Symposium On Biomedical Imaging (ISBI 2018)*. IEEE, 168–172.
- [149] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. 2018. The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data* 5, 1 (2018), 1–9.
- [150] Marc Combalia, Noel CF Codella, Veronica Rotemberg, Brian Helba, Veronica Vilaplana, Ofer Reiter, Cristina Carrera, Alicia Barreiro, Allan C Halpern, and Susana Puig. 2019. Bcn20000: Dermoscopic lesions in the wild. arXiv:1908.02288. Retrieved from <https://arxiv.org/abs/1908.02288>
- [151] Sema Candemir, Stefan Jaeger, Kannappan Palaniappan, Jonathan P Musco, Rahul K Singh, Zhiyun Xue, Alexandros Karargyris, Sameer Antani, George Thoma, and Clement J McDonald. 2013. Lung segmentation in chest radiographs using anatomical atlases with nonrigid registration. *IEEE Transactions on Medical Imaging* 33, 2 (2013), 577–590.
- [152] Stefan Jaeger, Alexandros Karargyris, Sema Candemir, Les Folio, Jenifer Siegelman, Fiona Callaghan, Zhiyun Xue, Kannappan Palaniappan, Rahul K Singh, and Sameer Antani. 2013. Automatic tuberculosis screening using chest radiographs. *IEEE Transactions on Medical Imaging* 33, 2 (2013), 233–245.
- [153] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. 1998. Gradient-based learning applied to document recognition. *Proceedings of the IEEE* 86, 11 (1998), 2278–2324.
- [154] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778.
- [155] Karen Simonyan and Andrew Zisserman. 2014. Very deep convolutional networks for large-scale image recognition. arXiv:1409.1556. Retrieved from <https://arxiv.org/abs/1409.1556>
- [156] Frank Rosenblatt. 1958. The perceptron: a probabilistic model for information storage and organization in the brain. *Psychological Review* 65, 6 (1958), 386.
- [157] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. 2017. ImageNet classification with deep convolutional neural networks. *Commun. ACM* 60, 6 (May 2017), 84–90. DOI : [10.1145/3065386](https://doi.org/10.1145/3065386)
- [158] Thomas F. Schranghamer, Aaryan Oberoi, and Saptarshi Das. 2020. Graphene memristive synapses for high precision neuromorphic computing. *Nature Communications* 11, 1 (2020), 5474.
- [159] Wolfgang Maass. 1997. Networks of spiking neurons: The third generation of neural network models. *Neural Networks* 10, 9 (1997), 1659–1671.
- [160] Charles R. Qi, Hao Su, Kaichun Mo, and Leonidas J. Guibas. 2017. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 652–660.

- [161] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. 2018. Mobilenetv2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 4510–4520.
- [162] Sachin Mehta and Mohammad Rastegari. 2021. Mobilevit: light-weight, general-purpose, and mobile-friendly vision transformer. arXiv:2110.02178. Retrieved from <https://arxiv.org/abs/2110.02178>
- [163] A Vaswani. 2017. Attention is all you need. *Advances in Neural Information Processing Systems*.
- [164] David W. Hosmer Jr, Stanley Lemeshow, and Rodney X. Sturdivant. 2013. *Applied Logistic Regression*. John Wiley & Sons.
- [165] Sepp Hochreiter, and Jürgen Schmidhuber. 1997. Long short-term memory. *Neural Comput.* 9, 8 (1997), 1735–1780.
- [166] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich. 2015. Going deeper with convolutions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 1–9.
- [167] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. 2017. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 4700–4708.
- [168] Jiayu Lin. 2016. *On the dirichlet distribution*. Department of Mathematics and Statistics, Queens University 10–11.
- [169] Anusha Lalitha, Shubhanshu Shekhar, Tara Javidi, and Farinaz Koushanfar. 2018. Fully decentralized federated learning. In *Third Workshop on Bayesian Deep Learning (NeurIPS)*.
- [170] Abhijit Guha Roy, Shayan Siddiqui, Sebastian Pölsterl, Nassir Navab, and Christian Wachinger. 2019. BrainTorrent: A Peer-to-Peer Environment for Decentralized Federated Learning.
- [171] David E. Rumelhart, Geoffrey E. Hinton, and Ronald J. Williams. 1985. Learning internal representations by error propagation. *Parallel Distributed Processing: Explorations in the Microstructure of Cognition*, Vol. 1. MIT Press. 318–362.
- [172] Ying Li, Xingwei Wang, Rongfei Zeng, Praveen Kumar Donta, Ilir Murturi, Min Huang, and Schahram Dustdar. 2023. Federated domain generalization: A survey. arXiv:2306.01334. Retrieved from <https://arxiv.org/abs/2306.01334>
- [173] Md Palash Uddin, Yong Xiang, Mahmudul Hasan, Jun Bai, Yao Zhao, and Longxiang Gao. 2025. A systematic literature review of robust federated learning: Issues, solutions, and future research directions. *ACM Computing Surveys*.
- [174] Wu-Chun Chung, Yan-Hui Lin, and Jau-Ai Luo. 2024. Ring-based decentralized federated learning with cosine similarity grouping. In *2024 International Conference on Consumer Electronics-Taiwan (ICCE-Taiwan)*, 113–114.
- [175] Mengli Wei, Wenwu Yu, and Duxin Chen. 2024. AccDFL: Accelerated decentralized federated learning for healthcare IoT networks. *IEEE Internet of Things Journal*.
- [176] Xinyi Fan, Jielei Chu, Zhaoyu Li, Jiangtao Hu, and Tianrui Li. 2024. Trust mechanisms in decentralized federated learning: Enhancing model parameter aggregation. In *2024 4th International Conference on Industrial Automation, Robotics and Control Engineering (IARCE)*, 295–301.
- [177] Chao Feng, Alberto Huertas Celdrán, Zien Zeng, Zi Ye, Jan von der Assen, G r me Bovet, and Burkhard Stiller. 2024. Leveraging MTD to mitigate poisoning attacks in decentralized FL with Non-IID data. In *2024 IEEE International Conference on Big Data (BigData)*, 7745–7754.
- [178] Yunming Liao, Yang Xu, Hongli Xu, Min Chen, Lun Wang, and Chunming Qiao. 2024. Asynchronous decentralized federated learning for heterogeneous devices. *IEEE/ACM Transactions on Networking*.
- [179] Zheyi Chen, Yunjing Ren, Yujie Xue, Alireza Jolfaei, Amr Tolba, Keping Yu, and Hailin Feng. 2025. A trustworthy decentralized federated learning framework for consumer electronics: Mitigating large-scale AIoT heterogeneity through transfer knowledge integration. *IEEE Transactions on Consumer Electronics* 71, 1 (2025), 1918–1925. DOI : [10.1109/TCE.2024.3472468](https://doi.org/10.1109/TCE.2024.3472468)
- [180] Huanhuan Ge, Shiva Raj Pokhrel, Zhenyu Liu, Jinlong Wang, and Gang Li. 2024. PFL-DKD: Modeling decoupled knowledge fusion with distillation for improving personalized federated learning. *Computer Networks* 254, 110758.
- [181] Behnaz Soltani, Venus Haghighi, Yipeng Zhou, Quan Z. Sheng, and Lina Yao. 2024. DFLStar: A decentralized federated learning framework with self-knowledge distillation and participant selection. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management*, 2108–2117.
- [182] Chao Feng, Alberto Huertas Celdrán, Zien Zeng, Zi Ye, Jan von der Assen, G r me Bovet, and Burkhard Stiller. 2024. Leveraging MTD to mitigate poisoning attacks in decentralized FL with Non-IID data. In *2024 IEEE International Conference on Big Data (BigData)*, 7745–7754.
- [183] Minghong Fang, Zifan Zhang, Hairi, Prashant Khanduri, Jia Liu, Songtao Lu, Yuchen Liu, and Neil Gong. 2024. Byzantine-robust decentralized federated learning. In *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security*, 2874–2888.
- [184] Qian Chen, Zilong Wang, Wenjing Zhang, and Xiaodong Lin. 2023. PPT: A privacy-preserving global model training protocol for federated learning in P2P networks. *Computers & Security* 124 (2023), 102966. DOI : [10.1016/j.cose.2022.102966](https://doi.org/10.1016/j.cose.2022.102966)

- [185] Shaik Karimullah, Ali H. Wheeb, and Fahimuddin Shaik. 2024. Detection and classification of brain tumor from MRI and CT images using harmony search optimization and deep learning. *Journal of Artificial Intelligence Research & Advances* 11, 3 (2024), 31–49.
- [186] Ying Li, Xingwei Wang, HaoDong Li, Praveen Kumar Donta, Min Huang, and Schahram Dustdar. 2025. Communication-Efficient federated learning for heterogeneous clients. *ACM Transactions on Internet Technology* 25, 2 (2025), 1–37.
- [187] Praveen Kumar Donta, Qiyang Zhang, and Schahram Dustdar. 2025. Performance measurements in the AI-Centric computing continuum systems. arXiv:2506.22884. Retrieved from <https://arxiv.org/abs/2506.22884>
- [188] Alaa Saleh, Roberto Morabito, Schahram Dustdar, Sasu Tarkoma, Susanna Pirttikangas, and Lauri Lovén. 2025. Towards message brokers for generative AI: Survey, challenges, and opportunities. *ACM Computing Surveys* 58, 1 (2025), 1–37.

A Appendix

While previous surveys have examined specific topics in CFL, this section offers a comprehensive analysis of the related surveys and categorizes the discussions into four major topics: security and privacy issues, inconsistent data distribution, the effects of optimization mechanisms, and the transition from CFL to DFL. Lastly, current studies on DFL and research gaps in this field are presented.

A.1 Security and Privacy

The 2021 survey [18] explored the security and privacy issues of FL, focusing on key technical aspects such as architecture, data partitioning, algorithms, and communication costs. The study identified vulnerabilities and potential threats in FL, analyzed the latest security risks, and reviewed existing defense technologies, offering insights into future research directions. In 2022, another study [19] highlighted critical security threats, including poisoning attacks, backdoor exploits, and GAN-based attacks. Through an in-depth analysis of attack patterns and defense techniques in FL, the research emphasized the potential of blockchain technology to enhance security. The study also summarized existing privacy vulnerabilities and proposed directions for addressing unresolved issues.

In 2025, a newly published SLR survey [173] comprehensively examined eight major themes for enhancing the robustness of FL. The research provided an in-depth analysis of various defense techniques and their limitations, and proposed future research directions such as hybrid defenses, ensemble learning, and adaptive mechanisms. Emphasizing the critical importance of FL robustness in countering adversarial attacks and noisy data, this comprehensive review offers valuable insights and a roadmap for both academic research and industrial applications.

A.2 Non-IID Data

Few studies have addressed the problem of Non-IID data distribution in FL. In 2021, a study [6] proposed methods to tackle this issue, as in-depth research on the impact of Non-IID data was lacking at the time. The study explored various types of data distributions, such as feature skew, label skew, temporal skew, and quantity skew, and analyzed challenges across different FL architectures. Solutions were categorized into data, algorithms, and systems, and unresolved issues like privacy protection and algorithmic differences were identified. A later study [21] provided a systematic overview of Non-IID data challenges in FL from 2016 to 2022. The study categorized research into four dimensions: data, models, algorithms, and frameworks, reviewing methods for handling Non-IID data distribution and outlining future trends in this area. The most recent research [7] focused on Non-IID data in FL architectures from 2022 to 2023, examining its effects on communication efficiency, model convergence, and accuracy. The study addressed communication and privacy challenges using data partitioning, client selection, differential privacy, and secure

aggregation, while also discussing applications. However, these studies mainly focused on traditional FL architectures and did not explore decentralized frameworks.

A.3 Optimization

In FL, the efficiency and performance of model training are highly dependent on the optimization of communication and computational resources. To address these challenges, a substantial body of literature has conducted systematic reviews on optimization issues in FL, with particular focus on key aspects such as communication cost, computational load, and client selection mechanisms. The following summarizes several important surveys and research works related to optimization in FL.

The research [22] focused on improving communication and computation efficiency in FL. While FL ensures data security, communication costs increase with more clients. Their work covered topics like client selection, update compression, network topology, wireless aggregation, and resource allocation, as well as hyperparameter optimization and model pruning. Another survey [23] concentrated on communication efficiency, especially heterogeneity and communication costs in training environments. The study classified these issues into three areas: communication efficiency, training environment, and resource allocation, analyzing methods like quantization, sparsification, and knowledge distillation, while summarizing the latest communication algorithms in FL.

Synchronous training is commonly used in traditional FL. However, in heterogeneous environments, especially with slower devices, model training efficiency can be compromised. Recent studies have addressed these issues, with research [24] explored computational heterogeneity, including cross-device variability, resource competition, and challenges related to communication and data size. The study also looked at neural network architecture differences and suggested solutions like client selection and asynchronous aggregation to handle device heterogeneity. Other research has proposed client selection mechanisms for heterogeneous FL. FedAvg often uses random client selection, which can reduce effectiveness due to variations in computational and communication capabilities and data distributions. The research [25] categorized client selection methods into trust-based, time-based, reinforcement-learning-based, and probability-based approaches. The study highlighted that client selection research is still developing and needs more focus on decentralized methods to avoid single points of failure.

Recent research in heterogeneous environments has focused on client selection in FL. In mobile networks, limited and unstable device communication bandwidth means only a subset of clients can be selected for model training. Therefore, an efficient client selection mechanism is crucial to improve overall FL training effectiveness. The previous survey [27] provided a detailed discussion on client selection techniques in mobile networks, highlighting key challenges in the selection process. Client selection faces challenges such as device and data heterogeneity, dynamics, trustworthiness, and fairness. Current methods include threshold-based, reputation-based, probability-based, reinforcement-learning-based, group-based, weight-divergence-based, and offloading-based selection. Despite these methods, many challenges in FL remain open and need further research. The latest survey [28] extensively reviewed client selection in FL, identifying critical issues. Prioritizing certain clients may lead to performance problems, so a balance between optimal and diverse client selection mechanisms is necessary. Future research directions include determining the optimal number of clients, performance guarantees, benchmarks, and scalability to other FL frameworks, along with identifying new research opportunities.

A.4 Decentralized Federated Learning

While research in FL has been extensive, inherent issues still remain, such as a single point of failure, security concerns, and communication bottlenecks. DFL has emerged as a technique to tackle

these issues. Current DFL surveys focused on integrating FL with blockchain [9–11, 20]. In particular, a survey [20] conducted an SLR on using blockchain to address security and privacy challenges in FL, exploring blockchain-based federated learning architectures and deployment frameworks. The study delved into communication efficiency, security assessment, and application scenarios. Despite effective progress in implementing blockchain for decentralized learning, certain challenges persist, including communication efficiency, malicious attacks, and data quality concerns.

The survey on asynchronous FL [26] analyzed challenges such as device and data heterogeneity, privacy, and security. The study reviewed application scenarios for various devices and suggested future research directions based on a comprehensive analysis. The latest surveys on DFL [8, 12] provided a statistical analysis of foundational aspects to cover architectures, network topologies, communication mechanisms, security methods, and key performance indicators. Additionally, these surveys examined optimization mechanisms used in DFL, compared relevant features, and outlined application scenarios for existing DFL frameworks.

A.5 Research Gap

Existing survey papers primarily explore FL issues related to privacy protection, data distribution, and communication efficiency. Multiple literature reviews have addressed security in centralized architectures, Non-IID data challenges, communication costs, heterogeneity, and client selection mechanisms. These studies mainly focus on CFL, with only a few publications beginning to examine issues within DFL frameworks, mostly combining blockchain with FL to enhance security and trust mechanisms. However, these studies on Non-IID problems predominantly concentrate on CFL architectures, lacking systematic organization and in-depth analysis of challenges and solutions for Non-IID data distribution in DFL systems. Previous review studies have proposed solutions from data, algorithmic, and system perspectives, analyzing Non-IID effects on model convergence, communication efficiency, and accuracy. Yet most discussions remain limited to centralized architectures, with insufficient integrated examination of data heterogeneity, network topology, optimization mechanisms, and security trust in decentralized environments.

Compared with previous surveys focusing on CFL or discussing only single aspects, this article presents the first systematic integration of research related to Non-IID data in DFL frameworks. Our work fills the gap in a comprehensive understanding of DFL systems under multiple challenges including data heterogeneity, network topology, optimization mechanisms, and security trust. The contribution helps the academic community grasp current DFL developments and future trends while providing concrete references and development directions for practical applications, highlighting the importance of this article in related fields.

Received 13 September 2024; revised 25 October 2025; accepted 7 November 2025